Robustness and Verification Limits of a Hybrid DWT–DCT–SVD Image Watermark: A 100-Image Evaluation with False-Positive Controls

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Abstract—Singular-value watermarking in a wavelet–cosine transform domain is widely used for ownership marking, yet reported evaluations typically use a handful of images and score only the correlation between the extracted and the owner's logo. We re-evaluate a classical hybrid DWT–DCT–SVD scheme on 100 unique 512×512 images (94 real, 6 labelled synthetic) under 14 attacks plus an unattacked reference (3000 seeded trials), with an embedding-strength sweep, false-positive controls, a comparison with an open-source blind block-based hybrid and two classical baselines, and an exploratory per-frame video test. The scheme is imperceptible (mean PSNR 36.0 dB, SSIM 0.992) and recovers the logo (NC ≥ 0.75) in 85.1% of trials: ≥ 95 % of images survive JPEG (Q ≥ 20), median filtering, sharpening, rescaling and mild noise, but only 72% survive 15° rotation, 44% histogram equalisation and 1% border cropping. Two findings qualify common practice. First, the robustness–imperceptibility trade-off in the embedding strength α is flat for JPEG and appears only under harsher attacks. Second, a different logo decoded with the owner's key still yields NC 0.975 (genuine pair 0.977), and 10%–13% of unrelated-image controls exceed the customary NC 0.75 threshold: a high NC does not show that the claimed watermark is present. The blind block-based hybrid achieves lower bit-error rates under most attacks but is 13× slower and weaker under rotation. Code, data manifest and all results accompany the paper.

Index Terms—*digital watermarking, DWT, DCT, SVD, robustness, false-positive problem, normalised correlation, PSNR, SSIM.*

I. INTRODUCTION

Images are copied at no cost, so owners embed invisible signatures that can later be extracted to support an ownership claim [1], [2]. Spatial-domain marks such as least-significant-bit (LSB) substitution are fragile [3], whereas transform-domain schemes place the mark in perceptually significant spectral components [4]. SVD-based schemes for ownership protection were introduced by Liu and Tan [5], combined with the wavelet transform by Ganic and Eskicioglu [6], and later cascaded with the DCT in hybrid DWT–DCT–SVD designs [7], [8].

Two gaps motivate this study. First, evaluations of such schemes usually rely on a few standard images and report one number, the normalised correlation (NC) with the owner's logo. Second, SVD schemes that store the logo's singular vectors in the key have long been criticised for false positives and ambiguity attacks [9], [10], but the effect is rarely quantified, so the common NC ≥ 0.75 rule is used without knowing its false-alarm rate.

Contributions. (1) A seeded, fully logged evaluation of a classical DWT–DCT–SVD scheme on 100 unique images with 14 attacks and an unattacked reference. (2) Control experiments that measure the false-positive problem and a metric analysis (a blank image already scores NC 0.36 against our logo; a mean-centred variant, ZNCC, is added). (3) An embedding-strength study showing that the trade-off depends on the attack, correcting a single-attack plot. (4) A measured comparison with an existing open-source blind block-based hybrid [11] and two classical baselines, and an exploratory per-frame video test.

II. RELATED WORK

Foundations. Cox *et al.* [4] argued that a watermark must reside in perceptually significant coefficients and be detected by correlation; Koch and Zhao [12] tie block-DCT embedding to the JPEG model; Hartung and Kutter [1] survey the field. The DWT [13] and DCT [14] underlie JPEG-family codecs [15]. **SVD schemes.** Liu and Tan [5] modify singular values for rightful-ownership protection; Ganic and Eskicioglu [6] embed in the wavelet sub-bands; Lai and Tsai [16] and later hybrids [7], [8] report improved robustness (we did not re-evaluate those claims). **Critique.** Zhang and Li [9] commented on [5], and Ting [10] studied ambiguity attacks on [6]: when the logo's singular vectors are part of the key, the detector can reconstruct the key owner's logo regardless of what was embedded. **Geometry and video.** Lin *et al.* [17] show that rotation, scale and translation defeat correlation detectors unless the detection domain is invariant; Hartung and Girod [18] treat compressed video. Learned watermarking [19], [20] is outside our scope.

III. METHOD

A. Embedding and extraction

A one-level Haar DWT splits the 512×512 host into four sub-bands; the 256×256 approximation LL is transformed by a global orthonormal 2-D DCT (not block-wise) and factorised by SVD, $A = U \cdot \mathrm{diag}(S) \cdot V^{T}$. The 64×64 binary logo W is resized to the sub-band size and factorised, $W = U_w \cdot \mathrm{diag}(S_w) \cdot V_w^{T}$. With embedding strength α (0.10 unless stated),

$$S' = S + \alpha \cdot S_w , \quad A' = U \cdot \mathrm{diag}(S') \cdot V^{T} \qquad (1)$$

followed by the inverse DCT, the inverse DWT and clipping to [0, 255]. The key is $K = \{S_orig, U_w, V_w^{T}, \alpha\}$. Given a (possibly attacked) image, the same transforms give singular values S_att, and

$$S_w' = (S_att - S_orig)/\alpha , \quad W' = U_w \cdot \mathrm{diag}(S_w') \cdot V_w^{T} \qquad (2)$$

The scheme is therefore **non-blind** (S_orig is derived from the unmarked host), and the logo's identity is carried by the key (U_w, V_w), not by the image — the structural reason for the false-positive behaviour measured in Section V-E. The embedding and extraction code is unchanged from the first implementation of this project; numerical results of that implementation were reproduced to within 0.004 NC.

B. Attacks and protocol

Fourteen attacks plus the unattacked reference: JPEG at quality 90/50/20; Gaussian noise ($\sigma^2 = 0.005$, 0.02 on [0,1]); salt-and-pepper (2 %); 3×3 median; Gaussian blur ($\sigma = 1.5$); unsharp-mask sharpening; rotation by 5° and 15°; border zeroing with 25 % of each dimension removed (43.75 % of the area); 0.5× down/up-scaling; and histogram equalisation. The watermarked image is rounded to 8 bit before attacking (primary protocol); attacking the floating-point image, as in the first implementation, changed mean NC by less than 0.001 except for equalisation (0.644 vs 0.638). All stochastic attacks are seeded.

C. Metrics

Imperceptibility is measured against the unwatermarked original by PSNR = $10 \log_{10}(255^2/\mathrm{MSE})$ and SSIM [21]. Recovery is measured by $\mathrm{NC} = \Sigma\, W \cdot W' / (\|W\| \|W'\|)$, by BER after thresholding at 127.5, and by ZNCC, the NC of the mean-centred images. For this logo a blank white image scores NC = 0.359 and a rank-1 approximation 0.868, so NC alone flatters bright outputs. A logo is called recovered if NC ≥ 0.75, a convention we keep for comparability with earlier work; we could not trace it to a primary source and estimate its false-alarm rate in Section V-E.

IV. EXPERIMENTAL SETUP

Data. 100 unique images (94 real, 6 synthetic controls: gradients, grating, noise, checkerboard) were drawn from open sample collections hosted on GitHub (OpenCV, scikit-image, three.js textures and others), converted to grey (ITU-R BT.601), centre-cropped and resized to 512×512; near-duplicates were removed by a difference hash. The set is a convenience sample covering animals, urban scenes, objects, textures, portraits, paintings, documents, astronomy and medical images with dark/bright and low/high-contrast cases; it is not a recognised benchmark, and per-image provenance, licences and rights notes are recorded in the metadata.

Experiments. E1 imperceptibility (100 images); E2 robustness (3000 trials = 100 images × 15 conditions × 2 protocols); E3 an α sweep over eight values and four attacks (4000 rows); E4 false-positive controls (1600 rows): unmarked and marked *other* images decoded with the key of image *i*, and the same host marked with a *different* logo decoded with the key of logo A. S1 compares the scheme, at equal PSNR and with an identical 4095-bit payload, with `blind-watermark` 0.4.4 [11] (blind, block-based DWT–DCT–SVD), an 8×8 block-DCT baseline and an LSB baseline on 30 random real images. V1 applies the scheme frame by frame to two real clips.

Statistics and reproducibility. Means carry 95 % bootstrap intervals, pass rates Wilson intervals, associations Spearman correlations. Two complete executions gave identical metrics (maximum absolute difference 0.0); 43 unit tests pass. Software: NumPy [22], SciPy [23], scikit-image [24], PyWavelets [25], Pillow, OpenCV and FFmpeg; four vCPUs, single-thread BLAS.

V. RESULTS AND DISCUSSION

A. Imperceptibility

The mean PSNR of the stored 8-bit watermarked image is 35.99 dB (range 35.5–43.8; none below 30 dB) and the mean SSIM is 0.992 (Fig. 1). PSNR is almost host-independent (82 of 100 images lie in 35.5–35.75 dB) because the added energy is set by the logo's singular values. SSIM exposes exceptions: the near-black astronomy image reaches 35.9 dB but SSIM 0.58, and three images fall below 0.98; PSNR alone would have hidden this.



Fig. 1. Original, watermarked and amplified difference (image_066, α = 0.10).

B. Robustness

TABLE I
Robustness over 100 images per condition (8-bit protocol).

Attack	mean NC (95 % CI)	≥0.75 (%)	ZNCC	BER
No Attack	0.977 (0.972–0.980)	100	0.974	0.009
JPEG Q=90	0.977 (0.972–0.979)	100	0.974	0.009
JPEG Q=50	0.975 (0.971–0.979)	100	0.972	0.009
JPEG Q=20	0.956 (0.932–0.975)	97	0.949	0.009
Gauss. Noise (var=0.005)	0.922 (0.884–0.952)	95	0.911	0.015
Gauss. Noise (var=0.02)	0.806 (0.755–0.852)	86	0.778	0.041
Salt & Pepper (2%)	0.928 (0.900–0.950)	97	0.919	0.018
Median Filter 3x3	0.967 (0.960–0.973)	100	0.964	0.008
Gauss. Blur (sigma=1.5)	0.913 (0.890–0.932)	96	0.897	0.022
Rotation 5 deg	0.880 (0.848–0.907)	92	0.860	0.037
Rotation 15 deg	0.776 (0.733–0.818)	72	0.739	0.067
Cropping 25%	0.022 (0.006–0.045)	1	-0.036	0.140
Scaling 0.5x	0.938 (0.916–0.953)	98	0.926	0.017
Hist. equalisation	0.644 (0.599–0.688)	44	0.615	0.126
Sharpening	0.966 (0.954–0.975)	99	0.963	0.009

1277 of 1500 trials (85.1%) reach NC ≥ 0.75 including the 100 unattacked references, as counted in earlier work; counting only the 14 attacks it is 84.1% (mean NC 0.834). ZNCC is about 0.02 lower than NC and yields 84.3% overall, so the conclusions are not an artefact of the brightness offset. Table I and Fig. 2 show strong robustness to compression and filtering and failure under cropping and equalisation. Rotation is weaker than sometimes assumed: mean NC 0.880 at 5° and 0.776 at 15° (92 % and 72 % pass). The scheme's robustness to JPEG stems from the SVD stage; the DCT is global and not aligned with JPEG's 8×8 blocks.

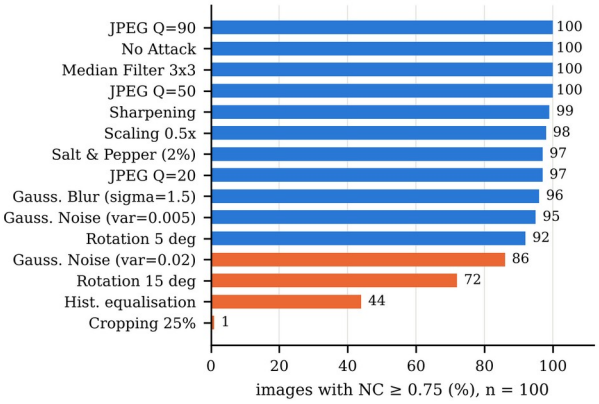


Fig. 2. Share of the 100 images with NC ≥ 0.75 per attack.

C. Which images fail

Robustness is image-dependent. NC after attack is negatively correlated with image contrast (Spearman ρ: salt-and-pepper -0.68, Gaussian noise σ^2=0.02 -0.65, 15° rotation -0.55); histogram equalisation instead tracks mean brightness (ρ = -0.31). Document-text images are the weakest category under noise (mean NC 0.53). These are associations with uncorrected p-values (n = 100), not demonstrated causes.

D. Embedding strength

Mean PSNR falls from 49.1 dB (α = 0.02) to 36.0 dB (0.10) and 24.2 dB (0.40). After JPEG Q = 50 the mean NC is essentially flat in α (0.960 at 0.02, 0.973 at 0.40). The trade-off appears under harsher attacks: for Gaussian noise (σ^2 = 0.02) the pass rate rises from 9% at α = 0.02 to 85% at 0.10 and 95% at 0.40 while PSNR drops below 30 dB for α ≥ 0.20 (Fig. 3). A single-attack plot on a second, zoomed NC axis — and a JPEG simulation that truncates floating-point pixels, which lowers NC only at α = 0.02 from 0.981 to 0.929 — had earlier suggested a steep trade-off for JPEG; we reproduced that figure exactly and traced the artefact.

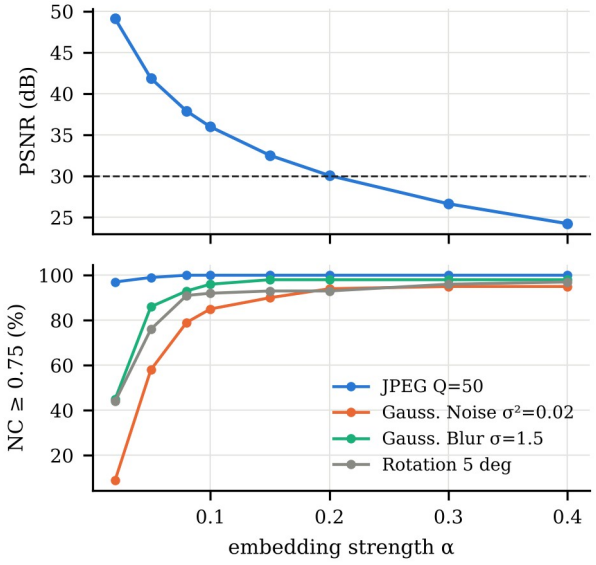


Fig. 3. Effect of α: PSNR (top) and pass rate after four attacks (bottom), 100 images.

E. False-positive controls

TABLE II
False-positive controls (E4).

	Control (NC against the owner's logo)	mean	p95	≥0.75 (%)
ref	genuine: own key, clean image	0.977	0.981	100.0
a	other image, unmarked, key of i	0.368	0.804	9.8
b	other image, marked, wrong key	0.401	0.832	12.6
c	other logo B, decoded with key A	0.975	0.980	100.0

Embedding a *different* logo in the same host and decoding with the key of logo A returns the owner's logo: NC 0.975, indistinguishable from the genuine 0.977 (Fig. 4, 200 trials) — the extractor reproduces the key owner's logo whatever singular values it finds. Unrelated images decoded with a wrong key score 0.37–0.40 on average, with 95th percentile 0.82; 9.8% (unmarked) and 12.6% (marked) exceed 0.75, but only about 1 % exceed 0.90. A detector calibrated to a 5 % false-alarm rate would reject 19% of the attacked genuine trials. Decoding the unmarked host with its own key gives NC = 0 by construction (S_att = S_orig) and must not be read as evidence. An attempt to cure the problem by embedding the logo's principal component made it worse (an unmarked image still decoded at NC 0.98), because the key then contains the whole logo. Claims of ownership therefore require an additional safeguard, for example an image-bound signature, which is future work.

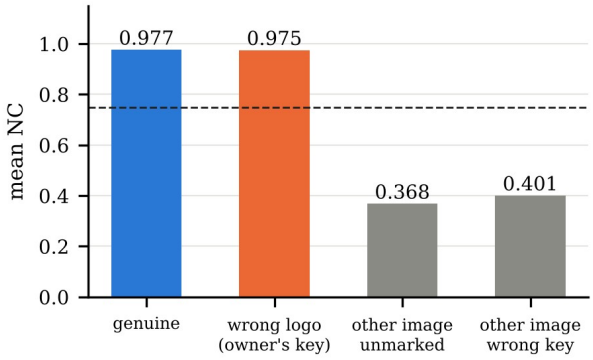


Fig. 4. Mean NC for the genuine pair and for the controls (dashed: 0.75).

F. Comparison with existing technology

TABLE III
Same 4095-bit payload and attacks.

BER (30 images)	Project	blind-wm.	block DCT	LSB
JPEG Q=50	0.010	0.004	0.296	0.426
JPEG Q=20	0.010	0.074	0.469	0.531
Gauss. noise	0.038	0.005	0.182	0.500
Rotation 5°	0.029	0.214	0.493	0.504
Cropping	0.135	0.060	0.124	0.128
Scaling 0.5×	0.015	0.004	0.483	0.455
PSNR (dB)	35.9	42.4	35.1	69.5
embed/extract (ms)	32/17	420/315	26/12	2/1

At equal PSNR the project is far more robust than the LSB and 8×8 block-DCT baselines, which approach chance (BER 0.5) under JPEG, rescaling and rotation (Fig. 5). It is, however, **not** superior to the existing blind block-based hybrid [11]: that package has lower BER under JPEG Q = 50 (0.004 vs 0.010), noise, blur, scaling and cropping at a higher PSNR (42.4 vs 35.9 dB) without a key, but is 13× slower to embed and 18× slower to extract, and worse under rotation (0.214 vs 0.029) and JPEG Q = 20 (0.074 vs 0.010). Its YUV design and our channel-wise attacks may favour it. Library-level checks

showed that DCT, SVD, Haar DWT and JPEG implementations in SciPy, OpenCV, NumPy and Pillow agree to $\leq 1.5 \cdot 10^{-16}$ (relative) or bit-identically, and that OpenCV is faster for attack operations (e.g. 3×3 median ≈ 340×, partly because it works on 8-bit data); the whole experiment runs in about 80 s on four cores, so speed was not a constraint.

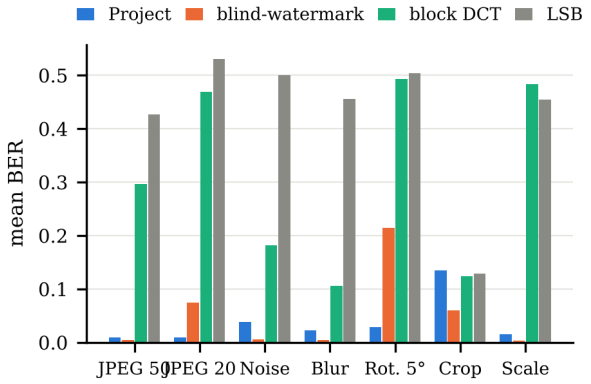


Fig. 5. Mean bit-error rate of four schemes (lower is better; 0.5 = chance).

G. Video feasibility

Applied frame by frame (each frame with its own key) to 60 frames of two real clips and re-encoded with H.264 (CRF 18/23/28) and MJPEG (q 5/15), the logo was recovered in all 720 decoded frames (NC ≥ 0.97); embedding takes about 32–41 ms per frame on one core. This is a feasibility check only: temporal attacks, colour video and hardware codecs were not tested, and a naive frame-drop test with position-aligned keys still gave NC 0.97 because neighbouring frames are nearly identical — a further symptom of key-dependence.

VI. LIMITATIONS

(i) The false-positive problem is measured, not solved. (ii) The scheme is non-blind and needs a key of about 2 MB per image. (iii) Border cropping and rotation beyond 5° defeat it; equalisation is only partly survived. (iv) Grey-scale only; clean extraction is not lossless (BER ≈ 0.9 % from logo resampling). (v) The 94 real images are a convenience sample from GitHub-hosted sets with mostly undocumented photograph rights, standardised to grey squares; category-level results rest on 1–14 images. (vi) Only one existing watermarking package was benchmarked; learned methods [19], [20] and GPU variants were not. (vii) Timing varies by 5–10 % between runs. (viii) The video test covers 120 distinct frames from two clips.

VII. CONCLUSION AND FUTURE WORK

A classical DWT–DCT–SVD watermark is imperceptible (PSNR 36.0 dB, SSIM 0.992) and survives everyday processing — JPEG, median filtering, sharpening, rescaling, mild noise — on at least 95 % of 100 images, but not cropping, equalisation or strong rotation. Larger-scale testing changes two common conclusions: the α trade-off depends on the attack, and a high normalised correlation does not prove that the claimed watermark is present, because a different logo decoded with the owner's key scores almost the same. An existing blind block-based hybrid outperforms the scheme on most attacks, so the scheme's value lies in its transparency and compactness, not in robustness. Future work: bind the logo to the image (signature of image features), block-based redundant embedding for cropping, geometric re-synchronisation [17], colour via luminance, and index-aware keys with temporal redundancy for video [18].

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APPENDIX A: SOURCE CODE

Python 3.13 · NumPy · SciPy · scikit-image · PyWavelets · Pillow · pandas · matplotlib (OpenCV and FFmpeg for the video test). Repository: github.com/krishangvikramsingh/claude-code, branch claude/elegant-tesla-c3k9fk, folder final_submission/source_code/src. The listing contains the algorithm, attacks, metrics, dataset builder, batch experiment, analysis and benchmark scripts; the data manifest (dataset_manifest.py) and the tests are in the repository.

A.1 watermark_core.py: Hybrid DWT-DCT-SVD embedding and extraction (algorithm core) (93 lines)

```
from __future__ import annotations

import numpy as np
import pywt
from scipy.fftpack import dct, idct
from skimage.transform import resize

_SUBBANDS = ("LL", "LH", "HL", "HH")


def dct2(block: np.ndarray) -> np.ndarray:
    return dct(dct(block.T, norm="ortho").T, norm="ortho")


def idct2(block: np.ndarray) -> np.ndarray:
    return idct(idct(block.T, norm="ortho").T, norm="ortho")


def _check_2d(name: str, a: np.ndarray, min_side: int = 8) -> np.ndarray:
    a = np.asarray(a)
    if a.ndim != 2:
        raise ValueError(f"{name} must be a 2-D greyscale array, got shape {a.shape}")
    if min(a.shape) < min_side:
        raise ValueError(f"{name} too small: {a.shape}")
    if not np.issubdtype(a.dtype, np.number):
        raise TypeError(f"{name} must be numeric, got {a.dtype}")
    if not np.isfinite(a).all():
        raise ValueError(f"{name} contains NaN/Inf")
    return a.astype(np.float64)


def embed_watermark(host: np.ndarray, watermark: np.ndarray, alpha: float = 0.10,
                    wavelet: str = "haar", subband: str = "LL"):
    host = _check_2d("host", host)
    watermark = _check_2d("watermark", watermark)
    if not alpha > 0:
        raise ValueError(f"alpha must be > 0, got {alpha}")
    if subband not in _SUBBANDS:
        raise ValueError(f"subband must be one of {_SUBBANDS}, got {subband!r}")

    LL, (LH, HL, HH) = pywt.dwt2(host, wavelet)
    subbands = {"LL": LL, "LH": LH, "HL": HL, "HH": HH}
    target = subbands[subband]

    wm_resized = resize(watermark, target.shape, anti_aliasing=True, preserve_range=True)

    target_dct = dct2(target)

    U, S, Vt = np.linalg.svd(target_dct, full_matrices=False)
    Uw, Sw, Vtw = np.linalg.svd(wm_resized, full_matrices=False)

    S_mod = S + alpha * Sw

    target_dct_mod = (U * S_mod) @ Vt
    target_mod = idct2(target_dct_mod)

    subbands[subband] = target_mod
    watermarked = pywt.idwt2((subbands["LL"], (subbands["LH"], subbands["HL"], subbands["HH"])), wavelet)

    watermarked = watermarked[:host.shape[0], :host.shape[1]]
    watermarked = np.clip(watermarked, 0, 255)

    key = {
        "U": U, "Vt": Vt, "S_original": S,
        "Uw": Uw, "Vtw": Vtw,
        "alpha": alpha, "wavelet": wavelet, "subband": subband,
        "wm_shape": watermark.shape, "target_shape": target.shape,
    }
    return watermarked, key


def extract_watermark(possibly_attacked: np.ndarray, key: dict) -> np.ndarray:
    img = _check_2d("image", possibly_attacked)
```

```
    wavelet = key["wavelet"]
    subband = key["subband"]

    LL, (LH, HL, HH) = pywt.dwt2(img, wavelet)
    subbands = {"LL": LL, "LH": LH, "HL": HL, "HH": HH}
    target = subbands[subband]

    target_shape = key["target_shape"]
    if target.shape != target_shape:
        target = resize(target, target_shape, anti_aliasing=True, preserve_range=True)

    target_dct = dct2(target)
    _, S_att, _ = np.linalg.svd(target_dct, full_matrices=False)

    Sw_extracted = (S_att - key["S_original"]) / key["alpha"]
    watermark_extracted = (key["Uw"] * Sw_extracted) @ key["Vtw"]

    watermark_extracted = resize(watermark_extracted, key["wm_shape"],
                                 anti_aliasing=True, preserve_range=True)
    return np.clip(watermark_extracted, 0, 255)
```

A.2 metrics.py: PSNR, SSIM, MSE, NC, ZNCC, BER (63 lines)

```
import numpy as np
from skimage.metrics import peak_signal_noise_ratio as _psnr
from skimage.metrics import structural_similarity as _ssim

PSNR_CAP_DB = 100.0


def _prep(a: np.ndarray, b: np.ndarray):
    a = np.clip(np.asarray(a, dtype=np.float64), 0, 255)
    b = np.clip(np.asarray(b, dtype=np.float64), 0, 255)
    if a.shape != b.shape:
        raise ValueError(f"shape mismatch: {a.shape} vs {b.shape}")
    return a, b


def mse(original: np.ndarray, test: np.ndarray) -> float:
    a, b = _prep(original, test)
    return float(np.mean((a - b) ** 2))


def psnr(original: np.ndarray, test: np.ndarray) -> float:
    a, b = _prep(original, test)
    if np.array_equal(a, b):
        return PSNR_CAP_DB
    return float(min(_psnr(a, b, data_range=255), PSNR_CAP_DB))


def ssim(original: np.ndarray, test: np.ndarray) -> float:
    a, b = _prep(original, test)
    return float(_ssim(a, b, data_range=255))


def normalized_correlation(original_wm: np.ndarray, extracted_wm: np.ndarray) -> float:
    a = np.asarray(original_wm, dtype=np.float64).flatten()
    b = np.asarray(extracted_wm, dtype=np.float64).flatten()
    if a.shape != b.shape:
        raise ValueError(f"shape mismatch: {original_wm.shape} vs {extracted_wm.shape}")
    num = np.sum(a * b)
    den = np.sqrt(np.sum(a ** 2)) * np.sqrt(np.sum(b ** 2))
    if den == 0:
        return 0.0
    return float(num / den)


def zero_normalized_correlation(original_wm: np.ndarray, extracted_wm: np.ndarray) -> float:
    a = np.asarray(original_wm, dtype=np.float64).flatten()
    b = np.asarray(extracted_wm, dtype=np.float64).flatten()
    if a.shape != b.shape:
        raise ValueError(f"shape mismatch: {original_wm.shape} vs {extracted_wm.shape}")
    a = a - a.mean()
    b = b - b.mean()
    den = np.sqrt(np.sum(a ** 2)) * np.sqrt(np.sum(b ** 2))
    if den == 0:
        return 0.0
    return float(np.sum(a * b) / den)


def bit_error_rate(original_wm: np.ndarray, extracted_wm: np.ndarray, thresh: float = 127.5) -> float:
    a = (np.asarray(original_wm, dtype=np.float64) > thresh).astype(np.uint8).flatten()
    b = (np.asarray(extracted_wm, dtype=np.float64) > thresh).astype(np.uint8).flatten()
```

```
    if a.shape != b.shape:
        raise ValueError(f"shape mismatch: {original_wm.shape} vs {extracted_wm.shape}")
    return float(np.mean(a != b))
```

A.3 attacks.py: The 14 attacks plus the unattacked reference, seeded (112 lines)

```
from __future__ import annotations

import io

import numpy as np
from PIL import Image
from scipy import ndimage
from skimage.exposure import equalize_hist
from skimage.transform import rotate, resize
from skimage.util import random_noise


def jpeg_compression(img: np.ndarray, quality: int = 50) -> np.ndarray:
    if not 1 <= quality <= 95:
        raise ValueError("quality must be in 1..95")
    im = Image.fromarray(np.clip(np.rint(img), 0, 255).astype(np.uint8))
    buf = io.BytesIO()
    im.save(buf, format="JPEG", quality=quality)
    buf.seek(0)
    return np.array(Image.open(buf).convert("L")).astype(np.float64)


def gaussian_noise(img: np.ndarray, var: float = 0.01, rng=None) -> np.ndarray:
    noisy = random_noise(img / 255.0, mode="gaussian", var=var, rng=rng)
    return noisy * 255.0


def salt_pepper_noise(img: np.ndarray, amount: float = 0.02, rng=None) -> np.ndarray:
    noisy = random_noise(img / 255.0, mode="s&p", amount=amount, rng=rng)
    return noisy * 255.0


def median_filter(img: np.ndarray, size: int = 3) -> np.ndarray:
    return ndimage.median_filter(img, size=size)


def gaussian_blur(img: np.ndarray, sigma: float = 1.5) -> np.ndarray:
    return ndimage.gaussian_filter(img, sigma=sigma)


def rotation_attack(img: np.ndarray, angle: float = 5.0) -> np.ndarray:
    rotated = rotate(img / 255.0, angle, resize=False, mode="edge")
    return rotated * 255.0


def cropping_attack(img: np.ndarray, frac: float = 0.25) -> np.ndarray:
    h, w = img.shape
    ch, cw = int(h * frac / 2), int(w * frac / 2)
    out = img.copy()
    out[:ch, :] = 0
    out[h - ch:, :] = 0
    out[:, :cw] = 0
    out[:, w - cw:] = 0
    return out


def scaling_attack(img: np.ndarray, factor: float = 0.5) -> np.ndarray:
    h, w = img.shape
    small = resize(img, (int(h * factor), int(w * factor)), anti_aliasing=True, preserve_range=True)
    back = resize(small, (h, w), anti_aliasing=True, preserve_range=True)
    return back


def histogram_equalization(img: np.ndarray) -> np.ndarray:
    eq = equalize_hist(img / 255.0)
    return eq * 255.0


def sharpening_attack(img: np.ndarray, amount: float = 1.0) -> np.ndarray:
    blurred = ndimage.gaussian_filter(img, sigma=1.0)
    sharpened = img + amount * (img - blurred)
    return np.clip(sharpened, 0, 255)


ATTACK_NAMES = [
    "No Attack", "JPEG Q=90", "JPEG Q=50", "JPEG Q=20",
    "Gaussian Noise (var=0.005)", "Gaussian Noise (var=0.02)", "Salt & Pepper (2%)",
```

```
    "Median Filter 3x3", "Gaussian Blur (sigma=1.5)", "Rotation 5 deg", "Rotation 15 deg",
    "Cropping 25%", "Scaling 0.5x", "Histogram Equalization", "Sharpening",
]

ATTACK_FAMILY = {
    "No Attack": "none", "JPEG Q=90": "compression", "JPEG Q=50": "compression", "JPEG Q=20": "compression",
    "Gaussian Noise (var=0.005)": "noise", "Gaussian Noise (var=0.02)": "noise", "Salt & Pepper (2%)": "noise",
    "Median Filter 3x3": "filtering", "Gaussian Blur (sigma=1.5)": "filtering", "Sharpening": "filtering",
    "Rotation 5 deg": "geometric", "Rotation 15 deg": "geometric", "Cropping 25%": "geometric",
    "Scaling 0.5x": "geometric", "Histogram Equalization": "global intensity",
}


def make_attack_suite(seed: int = 0) -> dict:
    return {
        "No Attack":                  lambda x: x.copy(),
        "JPEG Q=90":                  lambda x: jpeg_compression(x, 90),
        "JPEG Q=50":                  lambda x: jpeg_compression(x, 50),
        "JPEG Q=20":                  lambda x: jpeg_compression(x, 20),
        "Gaussian Noise (var=0.005)": lambda x: gaussian_noise(x, 0.005, rng=seed),
        "Gaussian Noise (var=0.02)":  lambda x: gaussian_noise(x, 0.02, rng=seed + 1),
        "Salt & Pepper (2%)":         lambda x: salt_pepper_noise(x, 0.02, rng=seed + 2),
        "Median Filter 3x3":          lambda x: median_filter(x, 3),
        "Gaussian Blur (sigma=1.5)":  lambda x: gaussian_blur(x, 1.5),
        "Rotation 5 deg":             lambda x: rotation_attack(x, 5),
        "Rotation 15 deg":            lambda x: rotation_attack(x, 15),
        "Cropping 25%":               lambda x: cropping_attack(x, 0.25),
        "Scaling 0.5x":               lambda x: scaling_attack(x, 0.5),
        "Histogram Equalization":     lambda x: histogram_equalization(x),
        "Sharpening":                 lambda x: sharpening_attack(x, 1.0),
    }


ATTACK_SUITE = make_attack_suite(0)
assert list(ATTACK_SUITE) == ATTACK_NAMES
```

A.4 logo.py: Procedural 64×64 binary watermark logo (17 lines)

```
import numpy as np
from PIL import Image, ImageDraw, ImageFont


def make_watermark_logo(size=64, text="CYBERION"):
    img = Image.new("L", (size * 4, size), color=0)
    draw = ImageDraw.Draw(img)
    try:
        font = ImageFont.load_default()
    except Exception:
        font = None
    draw.rectangle([0, 0, size * 4 - 1, size - 1], outline=255, width=3)
    draw.text((10, size // 2 - 6), text, fill=255, font=font)
    img = img.resize((size, size))
    arr = np.array(img).astype(np.float64)
    arr = (arr > 100).astype(np.float64) * 255.0
    return arr
```

A.5 image_io.py: Safe image loading and standardisation (86 lines)

```
from __future__ import annotations

import os
from dataclasses import dataclass

import numpy as np
from PIL import Image, UnidentifiedImageError

Image.MAX_IMAGE_PIXELS = 200_000_000


class ImageLoadError(RuntimeError):
    pass


@dataclass
class SourceInfo:
    width: int
    height: int
    mode: str
    fmt: str
    file_size: int


def _to_uint8_rgbish(im: Image.Image) -> Image.Image:
```

```
    mode = im.mode
    if mode in ("RGBA", "LA", "PA") or (mode == "P" and "transparency" in im.info):
        im = im.convert("RGBA")
        bg = Image.new("RGBA", im.size, (255, 255, 255, 255))
        return Image.alpha_composite(bg, im).convert("RGB")
    if mode in ("I;16", "I;16B", "I;16L", "I", "F"):
        arr = np.asarray(im, dtype=np.float64)
        lo, hi = float(arr.min()), float(arr.max())
        if hi <= lo:
            raise ImageLoadError("constant-valued high-bit-depth image")
        return Image.fromarray(((arr - lo) / (hi - lo) * 255.0).round().astype(np.uint8), "L")
    if mode == "1":
        return im.convert("L")
    if mode in ("L", "RGB"):
        return im
    return im.convert("RGB")


def load_gray_square(path: str, size: int = 512) -> tuple[np.ndarray, SourceInfo]:
    if not os.path.isfile(path):
        raise ImageLoadError(f"file not found: {path}")
    fsize = os.path.getsize(path)
    if fsize == 0:
        raise ImageLoadError(f"empty file: {path}")
    try:
        with Image.open(path) as im:
            fmt = im.format or ""
            w, h = im.size
            mode = im.mode
            im.load()
            rgb = _to_uint8_rgbish(im)
    except (UnidentifiedImageError, OSError, ValueError, SyntaxError) as exc:
        raise ImageLoadError(f"cannot decode {os.path.basename(path)}: {exc}") from exc
    if min(w, h) < 16:
        raise ImageLoadError(f"image too small ({w}x{h})")

    gray = rgb.convert("L")
    side = min(gray.size)
    left, top = (gray.width - side) // 2, (gray.height - side) // 2
    gray = gray.crop((left, top, left + side, top + side)).resize((size, size), Image.LANCZOS)
    arr = np.asarray(gray, dtype=np.float64)
    return arr, SourceInfo(w, h, mode, fmt, fsize)


def save_gray_png(arr: np.ndarray, path: str) -> None:
    os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
    Image.fromarray(np.clip(np.rint(arr), 0, 255).astype(np.uint8), "L").save(path)


def verify_png(path: str, expected_shape: tuple[int, int] | None = None) -> bool:
    try:
        with Image.open(path) as im:
            im.load()
            a = np.asarray(im)
    except Exception:
        return False
    if a.ndim != 2 or a.dtype != np.uint8:
        return False
    if expected_shape is not None and a.shape != tuple(expected_shape):
        return False
    return bool(np.isfinite(a).all())
```

A.6 run_experiment.py: Original two-image experiment (reproduces the first report) (152 lines)

```
import os
import json
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from PIL import Image
from skimage import data as skdata
from skimage.transform import resize

from watermark_core import embed_watermark, extract_watermark
from metrics import psnr, ssim, normalized_correlation, bit_error_rate
from attacks import ATTACK_SUITE
from logo import make_watermark_logo

RESULTS_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "..", "..", "results",
"legacy_two_image")
os.makedirs(RESULTS_DIR, exist_ok=True)
```

```
def load_host_images():
    hosts = {
        "cameraman": resize(skdata.camera().astype(np.float64), (512, 512), preserve_range=True),
        "astronaut_gray": resize(skdata.astronaut().mean(axis=2).astype(np.float64), (512, 512),
preserve_range=True),
    }
    return hosts


def save_image(arr, path):
    Image.fromarray(np.clip(arr, 0, 255).astype(np.uint8)).save(path)


def main():
    alpha = 0.10
    hosts = load_host_images()
    watermark = make_watermark_logo(size=64, text="CYBERION")
    save_image(watermark, os.path.join(RESULTS_DIR, "watermark_logo.png"))

    all_rows = []
    per_host_extracted = {}

    for host_name, host in hosts.items():
        watermarked, key = embed_watermark(host, watermark, alpha=alpha)
        save_image(host, os.path.join(RESULTS_DIR, f"{host_name}_original.png"))
        save_image(watermarked, os.path.join(RESULTS_DIR, f"{host_name}_watermarked.png"))

        imperceptibility_psnr = psnr(host, watermarked)
        imperceptibility_ssim = ssim(host, watermarked)

        extracted_examples = {}
        for attack_name, attack_fn in ATTACK_SUITE.items():
            attacked = attack_fn(watermarked)
            extracted = extract_watermark(attacked, key)
            nc = normalized_correlation(watermark, extracted)
            ber = bit_error_rate(watermark, extracted)

            all_rows.append({
                "Host Image": host_name,
                "Attack": attack_name,
                "Watermarked PSNR (dB)": round(imperceptibility_psnr, 3),
                "Watermarked SSIM": round(imperceptibility_ssim, 4),
                "Extracted NC": round(nc, 4),
                "Extracted BER": round(ber, 4),
            })
            extracted_examples[attack_name] = extracted

        per_host_extracted[host_name] = extracted_examples

    df = pd.DataFrame(all_rows)
    df.to_csv(os.path.join(RESULTS_DIR, "results_table.csv"), index=False)
    print(df.to_string(index=False))

    for host_name, host in hosts.items():
        watermarked = np.array(Image.open(os.path.join(RESULTS_DIR,
f"{host_name}_watermarked.png"))).astype(np.float64)
        diff = np.abs(host - watermarked)
        fig, axes = plt.subplots(1, 3, figsize=(12, 4.2))
        axes[0].imshow(host, cmap="gray"); axes[0].set_title("Original Host"); axes[0].axis("off")
        axes[1].imshow(watermarked, cmap="gray"); axes[1].set_title("Watermarked"); axes[1].axis("off")
        im = axes[2].imshow(diff, cmap="hot"); axes[2].set_title("Absolute Difference (x1)");
axes[2].axis("off")
        fig.colorbar(im, ax=axes[2], fraction=0.046)
        plt.tight_layout()
        plt.savefig(os.path.join(RESULTS_DIR, f"{host_name}_comparison.png"), dpi=150)
        plt.close(fig)

    gallery_attacks = ["No Attack", "JPEG Q=50", "JPEG Q=20", "Gaussian Noise (var=0.02)",
                       "Median Filter 3x3", "Rotation 15 deg", "Cropping 25%", "Scaling 0.5x"]
    fig, axes = plt.subplots(2, 4, figsize=(14, 7))
    for ax, aname in zip(axes.flat, gallery_attacks):
        ex = per_host_extracted["cameraman"][aname]
        ax.imshow(ex, cmap="gray")
        row = df[(df["Host Image"] == "cameraman") & (df["Attack"] == aname)].iloc[0]
        ax.set_title(f"{aname}\nNC={row['Extracted NC']:.3f}", fontsize=9)
        ax.axis("off")
    plt.tight_layout()
    plt.savefig(os.path.join(RESULTS_DIR, "extraction_gallery.png"), dpi=150)
    plt.close(fig)

    sub = df[df["Host Image"] == "cameraman"].sort_values("Extracted NC")
```

```
    fig, ax = plt.subplots(figsize=(9, 6))
    colors = ["#c0392b" if v < 0.75 else "#27ae60" for v in sub["Extracted NC"]]
    ax.barh(sub["Attack"], sub["Extracted NC"], color=colors)
    ax.axvline(0.75, color="black", linestyle="--", linewidth=1, label="Recognizability threshold (NC=0.75)")
    ax.set_xlabel("Normalized Correlation (NC)")
    ax.set_title("Watermark Robustness Across Attack Types (host: cameraman)")
    ax.legend(loc="lower right")
    plt.tight_layout()
    plt.savefig(os.path.join(RESULTS_DIR, "nc_bar_chart.png"), dpi=150)
    plt.close(fig)

    host = hosts["cameraman"]
    alphas = [0.02, 0.05, 0.08, 0.10, 0.15, 0.20, 0.30, 0.40]
    sweep_rows = []
    for a in alphas:
        wm_img, k = embed_watermark(host, watermark, alpha=a)
        p = psnr(host, wm_img)
        attacked = ATTACK_SUITE["JPEG Q=50"](wm_img)
        ext = extract_watermark(attacked, k)
        nc = normalized_correlation(watermark, ext)
        sweep_rows.append({"alpha": a, "PSNR": p, "NC_after_JPEG50": nc})
    sweep_df = pd.DataFrame(sweep_rows)
    sweep_df.to_csv(os.path.join(RESULTS_DIR, "alpha_sweep.csv"), index=False)

    fig, ax1 = plt.subplots(figsize=(8, 5.5))
    ax2 = ax1.twinx()
    ax1.plot(sweep_df["alpha"], sweep_df["PSNR"], "o-", color="#2980b9", label="PSNR (imperceptibility)")
    ax2.plot(sweep_df["alpha"], sweep_df["NC_after_JPEG50"], "s-", color="#e67e22", label="NC after JPEG Q=50
(robustness)")
    ax1.set_xlabel("Embedding strength (alpha)")
    ax1.set_ylabel("PSNR (dB)", color="#2980b9")
    ax2.set_ylabel("NC after JPEG-50 attack", color="#e67e22")
    ax1.set_title("Imperceptibility vs. Robustness Trade-off")
    fig.legend(loc="upper center", bbox_to_anchor=(0.5, 0.02), ncol=2)
    plt.tight_layout()
    plt.savefig(os.path.join(RESULTS_DIR, "alpha_tradeoff.png"), dpi=150, bbox_inches="tight")
    plt.close(fig)

    summary = {
        "alpha_used_main_experiment": alpha,
        "hosts": list(hosts.keys()),
        "num_attacks_tested": len(ATTACK_SUITE),
        "mean_watermarked_psnr_db": round(df["Watermarked PSNR (dB)"].mean(), 3),
        "mean_watermarked_ssim": round(df["Watermarked SSIM"].mean(), 4),
        "mean_nc_all_attacks": round(df["Extracted NC"].mean(), 4),
        "attacks_with_NC_above_0.75": int((df["Extracted NC"] > 0.75).sum()),
        "attacks_total_rows": len(df),
    }
    with open(os.path.join(RESULTS_DIR, "summary.json"), "w") as f:
        json.dump(summary, f, indent=2)
    print("\nSUMMARY:", json.dumps(summary, indent=2))


if __name__ == "__main__":
    main()
```

A.7 build_dataset.py: Reproducible 100-image dataset builder (209 lines)

```
from __future__ import annotations

import argparse
import csv
import hashlib
import json
import logging
import os
import sys
import time
import urllib.parse
import urllib.request

import numpy as np
from PIL import Image
from scipy import ndimage

from dataset_manifest import REAL_IMAGES
from image_io import ImageLoadError, load_gray_square, save_gray_png

LOG = logging.getLogger("build_dataset")
RAW = "https://raw.githubusercontent.com/"
SIZE = 512
```

```
def sha256_file(path: str) -> str:
    h = hashlib.sha256()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(1 << 20), b""):
            h.update(chunk)
    return h.hexdigest()


def dhash64(arr: np.ndarray) -> int:
    im = Image.fromarray(np.clip(arr, 0, 255).astype(np.uint8)).resize((9, 8), Image.LANCZOS)
    a = np.asarray(im, dtype=np.int16)
    bits = (a[:, 1:] > a[:, :-1]).flatten()
    return int("".join("1" if b else "0" for b in bits), 2)


def hamming(a: int, b: int) -> int:
    return bin(a ^ b).count("1")


def download(rel: str, dest: str, retries: int = 3) -> bool:
    url = RAW + rel
    last = None
    for k in range(retries):
        try:
            with urllib.request.urlopen(urllib.request.Request(url, headers={"User-Agent": "ivp-watermark-
build/1.0"}), timeout=60) as r:
                data = r.read()
            if not data:
                raise IOError("empty response")
            with open(dest, "wb") as f:
                f.write(data)
            return True
        except Exception as exc:
            last = exc
            time.sleep(1.5 * (k + 1))
    LOG.warning("download failed %s (%s)", url, last)
    return False


def fetch_scipy(name: str, dest: str) -> bool:
    try:
        import scipy.datasets as sd
        arr = getattr(sd, name)()
        Image.fromarray(arr).save(dest)
        return True
    except Exception as exc:
        LOG.warning("scipy.datasets.%s unavailable: %s", name, exc)
        return False


def _smooth_noise(rng, sigma, mean, std):
    n = ndimage.gaussian_filter(rng.standard_normal((SIZE, SIZE)), sigma)
    n = (n - n.mean()) / n.std()
    return np.clip(mean + std * n, 0, 255)


def synth_images():
    rng = np.random.default_rng(20260713)
    yy, xx = np.mgrid[0:SIZE, 0:SIZE].astype(np.float64)
    out = []
    out.append(("syn_hgradient", "horizontal linear gradient 0-255", np.tile(np.linspace(0, 255, SIZE), (SIZE,
1))))
    r = np.hypot(xx - SIZE / 2, yy - SIZE / 2)
    out.append(("syn_radial", "radial gradient (smooth, low detail)", np.clip(255 - r * 0.7, 0, 255)))
    out.append(("syn_grating", "2-D sinusoidal grating (periodic texture)", 128 + 100 * np.sin(2 * np.pi * (xx
* 9 + yy * 5) / SIZE)))
    out.append(("syn_white_noise", "white Gaussian noise, mean 128 / std 40", np.clip(128 + 40 *
rng.standard_normal((SIZE, SIZE)), 0, 255)))
    out.append(("syn_lowcontrast", "smooth noise, mean 128 / std 8 (very low contrast)", _smooth_noise(rng, 6,
128, 8)))
    out.append(("syn_checker", "32-px checkerboard (hard edges)", 255.0 * (((xx // 32) + (yy // 32)) % 2)))
    out.append(("syn_dark", "smooth noise, mean 25 / std 12 (dark)", _smooth_noise(rng, 5, 25, 12)))
    out.append(("syn_bright", "smooth noise, mean 230 / std 12 (bright)", _smooth_noise(rng, 5, 230, 12)))
    f = np.fft.fftfreq(SIZE)
    fx, fy = np.meshgrid(f, f)
    amp = 1.0 / np.maximum(np.hypot(fx, fy), 1.0 / SIZE)
    spec = amp * np.exp(2j * np.pi * rng.random((SIZE, SIZE)))
    frac = np.real(np.fft.ifft2(spec))
    frac = (frac - frac.mean()) / frac.std()
    out.append(("syn_fractal", "1/f fractal noise texture", np.clip(128 + 45 * frac, 0, 255)))
    scene = np.full((SIZE, SIZE), 110.0)
    for _ in range(14):
        x0, y0 = rng.integers(0, SIZE - 80, 2)
```

```
        w, h = rng.integers(40, 160, 2)
        scene[y0:y0 + h, x0:x0 + w] = rng.integers(20, 240)
    out.append(("syn_blocks", "random grey rectangles on grey background", scene))
    return out


def main():
    ap = argparse.ArgumentParser()
    here = os.path.dirname(os.path.abspath(__file__))
    ap.add_argument("--out", default=os.path.join(here, "..", "..", "data"))
    ap.add_argument("--target", type=int, default=100)
    ap.add_argument("--dup-threshold", type=int, default=6)
    args = ap.parse_args()
    out = os.path.abspath(args.out)
    src_dir = os.path.join(out, "source_images")
    orig_dir = os.path.join(out, "original")
    os.makedirs(src_dir, exist_ok=True)
    os.makedirs(orig_dir, exist_ok=True)
    logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")

    rows, hashes, skipped = [], [], []
    for key, rel, group, lic, note, category in REAL_IMAGES:
        ext = os.path.splitext(urllib.parse.unquote(rel))[1] or ".png"
        if rel.startswith("scipy:"):
            dest = os.path.join(src_dir, f"{key}.png")
            ok = os.path.isfile(dest) or fetch_scipy(rel.split(":", 1)[1], dest)
            url = f"scipy.datasets.{rel.split(':', 1)[1]}()"
        else:
            dest = os.path.join(src_dir, f"{key}{ext}")
            ok = os.path.isfile(dest) and os.path.getsize(dest) > 0 or download(rel, dest)
            url = RAW + rel
        if not ok:
            skipped.append((key, "download failed"))
            continue
        try:
            arr, info = load_gray_square(dest, SIZE)
        except ImageLoadError as exc:
            skipped.append((key, str(exc)))
            LOG.warning("skip %s: %s", key, exc)
            continue
        if min(info.width, info.height) < 240:
            skipped.append((key, f"too small ({info.width}x{info.height})"))
            continue
        hsh = dhash64(arr)
        dup = next((k for k, h in hashes if hamming(h, hsh) <= args.dup_threshold), None)
        if dup:
            skipped.append((key, f"near-duplicate of {dup}"))
            LOG.info("duplicate %s ~ %s", key, dup)
            continue
        hashes.append((key, hsh))
        rows.append(dict(key=key, arr=arr, url=url, group=group, lic=lic, note=note,
                         category=category, info=info, sha=sha256_file(dest),
                         fname=os.path.basename(dest), synthetic=False))

    need = args.target - len(rows)
    if need < 0:
        rows = rows[:args.target]
        need = 0
    synth = synth_images()
    if need > len(synth):
        LOG.warning("only %d synthetic generators available; dataset will have %d images",
                    len(synth), len(rows) + len(synth))
    for key, desc, arr in synth[:need]:
        rows.append(dict(key=key, arr=arr, url="generated by build_dataset.py (seeded)", group="synthetic
control",
                         lic="n/a (generated)", note=desc, category="synthetic_control",
                         info=None, sha="", fname="", synthetic=True))

    meta_path = os.path.join(out, "metadata.csv")
    cols = ["image_id", "key", "original_filename", "source_group", "source_url", "repository_license",
            "rights_note", "original_width", "original_height", "original_mode", "original_format",
            "original_file_size_bytes", "source_sha256", "standardized_shape", "standardized_channels",
            "category", "is_synthetic", "mean_intensity", "contrast_std", "brightness_class",
            "contrast_class", "applicable_operations"]
    with open(meta_path, "w", newline="", encoding="utf-8") as f:
        w = csv.DictWriter(f, fieldnames=cols)
        w.writeheader()
        for i, r in enumerate(rows, 1):
            iid = f"image_{i:03d}"
            save_gray_png(r["arr"], os.path.join(orig_dir, iid + ".png"))
            a = np.rint(r["arr"])
            mean, std = float(a.mean()), float(a.std())
            info = r["info"]
```

```
            w.writerow(dict(
                image_id=iid, key=r["key"], original_filename=r["fname"], source_group=r["group"],
                source_url=r["url"], repository_license=r["lic"], rights_note=r["note"],
                original_width=info.width if info else SIZE, original_height=info.height if info else SIZE,
                original_mode=info.mode if info else "generated", original_format=info.fmt if info else "PNG",
                original_file_size_bytes=info.file_size if info else "", source_sha256=r["sha"],
                standardized_shape=f"{SIZE}x{SIZE}", standardized_channels=1, category=r["category"],
                is_synthetic=int(r["synthetic"]), mean_intensity=round(mean, 2), contrast_std=round(std, 2),
                brightness_class="dark" if mean < 80 else ("bright" if mean > 170 else "mid"),
                contrast_class="low" if std < 35 else ("high" if std > 65 else "medium"),
                applicable_operations="embed;extract;14 attacks + reference;PSNR;SSIM;NC;BER;alpha-
sweep;controls"))

    with open(os.path.join(out, "build_log.json"), "w") as f:
        json.dump({"images_written": len(rows), "real": sum(not r["synthetic"] for r in rows),
                   "synthetic": sum(r["synthetic"] for r in rows), "skipped": skipped}, f, indent=2)
    LOG.info("wrote %d images (%d real, %d synthetic); skipped %d",
             len(rows), sum(not r["synthetic"] for r in rows), sum(r["synthetic"] for r in rows), len(skipped))
    for k, why in skipped:
        LOG.info("  skipped %-16s %s", k, why)
    return 0 if len(rows) == args.target else 1


if __name__ == "__main__":
    sys.exit(main())
```

A.8 run_batch_experiment.py: 100-image batch experiment (E1-E4) (258 lines)

```
from __future__ import annotations

import os

for _v in ("OPENBLAS_NUM_THREADS", "OMP_NUM_THREADS", "MKL_NUM_THREADS"):
    os.environ.setdefault(_v, "1")

import argparse
import concurrent.futures as cf
import json
import logging
import platform
import resource
import sys
import time

import numpy as np
import pandas as pd

from attacks import ATTACK_FAMILY, ATTACK_NAMES, make_attack_suite
from image_io import save_gray_png, verify_png
from logo import make_watermark_logo
from metrics import bit_error_rate, normalized_correlation, psnr, ssim, zero_normalized_correlation
from watermark_core import embed_watermark, extract_watermark

HERE = os.path.dirname(os.path.abspath(__file__))
LOG = logging.getLogger("batch")

DEFAULT_CONFIG = {
    "alpha": 0.10,
    "image_size": 512,
    "logo_size": 64,
    "logo_text": "CYBERION",
    "other_logos": ["OTHERLOGO", "OWNER-B"],
    "protocols": ["uint8", "float"],
    "primary_protocol": "uint8",
    "base_seed": 20260713,
    "alphas": [0.02, 0.05, 0.08, 0.10, 0.15, 0.20, 0.30, 0.40],
    "sweep_attacks": ["JPEG Q=50", "Gaussian Noise (var=0.02)", "Gaussian Blur (sigma=1.5)", "Rotation 5 deg"],
    "n_controls_per_image": 5,
    "nc_threshold": 0.75,
    "data_dir": "../../data",
    "results_dir": "../../results",
    "outputs_dir": "../../outputs",
}


def _t():
    return time.perf_counter()


def _load_host(data_dir, image_id):
    from PIL import Image
    a = np.asarray(Image.open(os.path.join(data_dir, "original", image_id + ".png")), dtype=np.float64)
    if a.ndim != 2:
```

```
        raise ValueError(f"{image_id}: expected greyscale, got shape {a.shape}")
    return a


def _quantize(a):
    return np.clip(np.rint(a), 0, 255)


def process_image(task):
    idx, image_id, cfg, others = task
    data_dir = cfg["data_dir"]
    rows_att, rows_ctl, rows_sw = [], [], []
    rec = {"image_id": image_id, "status": "ok", "error": ""}
    try:
        host = _load_host(data_dir, image_id)
        logo = make_watermark_logo(cfg["logo_size"], cfg["logo_text"])
        alpha = cfg["alpha"]

        t0 = _t()
        wm_float, key = embed_watermark(host, logo, alpha=alpha)
        rec["t_embed_ms"] = (_t() - t0) * 1e3
        wm_u8 = _quantize(wm_float)
        out_path = os.path.join(cfg["outputs_dir"], "watermarked", f"{image_id}_wm.png")
        save_gray_png(wm_u8, out_path)
        rec["output_valid"] = bool(verify_png(out_path, host.shape))
        rec["psnr_float_db"] = psnr(host, wm_float)
        rec["ssim_float"] = ssim(host, wm_float)
        rec["psnr_db"] = psnr(host, wm_u8)
        rec["ssim"] = ssim(host, wm_u8)
        rec["max_abs_diff"] = float(np.abs(host - wm_u8).max())
        rec["mean_abs_diff"] = float(np.abs(host - wm_u8).mean())
        rec["key_bytes"] = int(sum(v.nbytes for v in key.values() if isinstance(v, np.ndarray)))

        suite = make_attack_suite(cfg["base_seed"] + 100 * idx)
        for protocol in cfg["protocols"]:
            base = wm_u8 if protocol == "uint8" else wm_float
            for name in ATTACK_NAMES:
                row = {"image_id": image_id, "protocol": protocol, "attack": name,
                       "family": ATTACK_FAMILY[name], "status": "ok", "error": ""}
                try:
                    t0 = _t()
                    attacked = suite[name](base)
                    row["t_attack_ms"] = (_t() - t0) * 1e3
                    t0 = _t()
                    ext = extract_watermark(attacked, key)
                    row["t_extract_ms"] = (_t() - t0) * 1e3
                    row["nc"] = normalized_correlation(logo, ext)
                    row["zncc"] = zero_normalized_correlation(logo, ext)
                    row["ber"] = bit_error_rate(logo, ext)
                except Exception as exc:
                    row.update(status="error", error=f"{type(exc).__name__}: {exc}")
                rows_att.append(row)

        sweep_suite = make_attack_suite(cfg["base_seed"] + 100 * idx + 50)
        for a in cfg["alphas"]:
            try:
                w, k = embed_watermark(host, logo, alpha=a)
                wq = _quantize(w)
                p_a, s_a = psnr(host, wq), ssim(host, wq)
            except Exception as exc:
                rows_sw.append({"image_id": image_id, "alpha": a, "attack": "(embed)", "status": "error",
                                "error": f"{type(exc).__name__}: {exc}"})
                continue
            for an in ["No Attack"] + list(cfg["sweep_attacks"]):
                row = {"image_id": image_id, "alpha": a, "attack": an, "psnr_db": p_a, "ssim": s_a,
                       "status": "ok", "error": ""}
                try:
                    ext = extract_watermark(sweep_suite[an](wq), k)
                    row["nc"] = normalized_correlation(logo, ext)
                    row["zncc"] = zero_normalized_correlation(logo, ext)
                    row["ber"] = bit_error_rate(logo, ext)
                except Exception as exc:
                    row.update(status="error", error=f"{type(exc).__name__}: {exc}")
                rows_sw.append(row)

        def ctl(kind, other, ncv, berv, zz, extra=""):
            rows_ctl.append({"image_id": image_id, "control": kind, "other": other,
                             "nc": ncv, "ber": berv, "zncc": zz, "note": extra})

        ext = extract_watermark(wm_u8, key)
        ctl("genuine (own key, no attack)", image_id, normalized_correlation(logo, ext), bit_error_rate(logo,
ext), zero_normalized_correlation(logo, ext))
        ext = extract_watermark(host, key)
```

```
        ctl("d: own host unwatermarked, own key", image_id, normalized_correlation(logo, ext),
bit_error_rate(logo, ext), zero_normalized_correlation(logo, ext),
            extra="degenerate: S_att == S_original so Sw' = 0")
        for oid in others:
            oh = _load_host(data_dir, oid)
            ext = extract_watermark(oh, key)
            ctl("a: other host unwatermarked, key_i", oid, normalized_correlation(logo, ext),
bit_error_rate(logo, ext), zero_normalized_correlation(logo, ext))
            ow, _ = embed_watermark(oh, logo, alpha=alpha)
            ext = extract_watermark(_quantize(ow), key)
            ctl("b: other host watermarked, key_i", oid, normalized_correlation(logo, ext),
bit_error_rate(logo, ext), zero_normalized_correlation(logo, ext))
        for text in cfg["other_logos"]:
            logo_b = make_watermark_logo(cfg["logo_size"], text)
            wb, kb = embed_watermark(host, logo_b, alpha=alpha)
            wbq = _quantize(wb)
            ext_a = extract_watermark(wbq, key)
            ext_b = extract_watermark(wbq, kb)
            ctl("c: same host, other logo B, decoded with key A (NC vs A)", text,
                normalized_correlation(logo, ext_a), bit_error_rate(logo, ext_a),
zero_normalized_correlation(logo, ext_a))
            ctl("c-ref: same host, logo B, decoded with key B (NC vs B)", text,
                normalized_correlation(logo_b, ext_b), bit_error_rate(logo_b, ext_b),
zero_normalized_correlation(logo_b, ext_b))
    except Exception as exc:
        rec.update(status="error", error=f"{type(exc).__name__}: {exc}")
        LOG.exception("image %s failed", image_id)
    rec["peak_rss_mb"] = resource.getrusage(resource.RUSAGE_SELF).ru_maxrss / 1024.0
    return rec, rows_att, rows_sw, rows_ctl


def env_info():
    import PIL, pywt, scipy, skimage
    cpu = ""
    try:
        with open("/proc/cpuinfo") as f:
            cpu = next(l.split(":", 1)[1].strip() for l in f if l.startswith("model name"))
    except Exception:
        pass
    return {"python": sys.version.split()[0], "platform": platform.platform(), "cpu": cpu,
            "cpu_count": os.cpu_count(), "numpy": np.__version__, "scipy": scipy.__version__,
            "scikit-image": skimage.__version__, "PyWavelets": pywt.__version__, "Pillow": PIL.__version__,
            "pandas": pd.__version__}


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--config", default=None)
    ap.add_argument("--workers", type=int, default=os.cpu_count() or 1)
    ap.add_argument("--limit", type=int, default=None, help="process only the first N images (debug)")
    args = ap.parse_args()
    logging.basicConfig(level=logging.INFO, format="%(asctime)s %(levelname)s %(message)s")

    cfg = dict(DEFAULT_CONFIG)
    if args.config:
        with open(args.config) as f:
            cfg.update(json.load(f))
    for k in ("data_dir", "results_dir", "outputs_dir"):
        cfg[k] = os.path.abspath(os.path.join(HERE, cfg[k]))
    os.makedirs(cfg["results_dir"], exist_ok=True)
    os.makedirs(os.path.join(cfg["outputs_dir"], "watermarked"), exist_ok=True)

    meta = pd.read_csv(os.path.join(cfg["data_dir"], "metadata.csv"))
    if args.limit:
        meta = meta.head(args.limit)
    ids = list(meta["image_id"])
    n = len(ids)
    rng = np.random.default_rng(cfg["base_seed"])
    tasks = []
    for i, iid in enumerate(ids):
        pool = [x for x in ids if x != iid]
        others = list(rng.choice(pool, size=min(cfg["n_controls_per_image"], len(pool)), replace=False))
        tasks.append((i, iid, cfg, others))

    LOG.info("processing %d images with %d worker(s)", n, args.workers)
    t_start = _t()
    results = []
    with cf.ProcessPoolExecutor(max_workers=args.workers) as ex:
        for res in ex.map(process_image, tasks):
            results.append(res)
            if len(results) % 10 == 0:
                LOG.info("  %d/%d images done", len(results), n)
    wall = _t() - t_start
```

```
    img_df = pd.DataFrame([r[0] for r in results]).sort_values("image_id")
    att_df = pd.DataFrame([row for r in results for row in r[1]])
    sw_df = pd.DataFrame([row for r in results for row in r[2]])
    ctl_df = pd.DataFrame([row for r in results for row in r[3]])
    rd = cfg["results_dir"]
    img_df.to_csv(os.path.join(rd, "results_images.csv"), index=False)
    att_df.to_csv(os.path.join(rd, "results_attacks.csv"), index=False)
    sw_df.to_csv(os.path.join(rd, "results_alpha_sweep.csv"), index=False)
    ctl_df.to_csv(os.path.join(rd, "results_controls.csv"), index=False)

    full = pd.read_csv(os.path.join(cfg["data_dir"], "metadata.csv"))
    st = img_df.set_index("image_id")["status"]
    att_ok = att_df.groupby("image_id")["status"].apply(lambda s: int((s == "ok").sum()))
    full["processing_status"] = full["image_id"].map(st).fillna(full.get("processing_status", "not processed"))
    full["attack_trials_ok"] = full["image_id"].map(att_ok).fillna(full.get("attack_trials_ok"))
    full["attack_trials_expected"] = len(ATTACK_NAMES) * len(cfg["protocols"])
    full.to_csv(os.path.join(cfg["data_dir"], "metadata.csv"), index=False)

    summary = {
        "config": {k: v for k, v in cfg.items() if not k.endswith("_dir")},
        "environment": env_info(),
        "images_requested": n,
        "images_ok": int((img_df["status"] == "ok").sum()),
        "watermarked_files_valid": int(img_df["output_valid"].fillna(False).sum()),
        "attack_trials_total": int(len(att_df)),
        "attack_trials_ok": int((att_df["status"] == "ok").sum()),
        "attack_trials_failed": int((att_df["status"] != "ok").sum()),
        "alpha_sweep_rows_ok": int((sw_df["status"] == "ok").sum()),
        "alpha_sweep_rows_total": int(len(sw_df)),
        "control_rows": int(len(ctl_df)),
        "wall_time_s": round(wall, 1),
        "workers": args.workers,
        "peak_rss_mb_max_over_workers": round(float(img_df["peak_rss_mb"].max()), 1),
    }
    with open(os.path.join(rd, "run_summary.json"), "w") as f:
        json.dump(summary, f, indent=2)
    LOG.info("summary: %s", json.dumps({k: v for k, v in summary.items() if k not in ("config",
"environment")}))
    return 0 if summary["attack_trials_failed"] == 0 and summary["images_ok"] == n else 1


if __name__ == "__main__":
    sys.exit(main())
```

A.9 analysis.py: Statistics, tables and figures from the recorded results (379 lines)

```
from __future__ import annotations

import os
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")

import json
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from PIL import Image
from scipy import stats

from attacks import ATTACK_FAMILY, ATTACK_NAMES, make_attack_suite
from logo import make_watermark_logo
from metrics import normalized_correlation, psnr, ssim, zero_normalized_correlation
from watermark_core import embed_watermark, extract_watermark

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, "..", ".."))
RES = os.path.join(ROOT, "results")
TAB = os.path.join(RES, "tables")
VIS = os.path.join(ROOT, "visualizations")
DATA = os.path.join(ROOT, "data")
os.makedirs(TAB, exist_ok=True)
os.makedirs(VIS, exist_ok=True)

THR = 0.75
BLUE, ORANGE, AQUA, GREY, INK = "#2a78d6", "#eb6834", "#1baf7a", "#8a8a85", "#222222"
plt.rcParams.update({
    "font.size": 10, "axes.titlesize": 11, "axes.labelsize": 10, "axes.edgecolor": "#888888",
    "axes.spines.top": False, "axes.spines.right": False, "axes.grid": True, "grid.color": "#e4e4e0",
    "grid.linewidth": 0.7, "axes.axisbelow": True, "figure.facecolor": "white", "axes.facecolor": "white",
    "savefig.dpi": 200, "savefig.bbox": "tight", "legend.frameon": False,
})
```

```
rng = np.random.default_rng(20260713)


def wilson(k, n, z=1.96):
    if n == 0:
        return (np.nan, np.nan)
    p = k / n
    d = 1 + z * z / n
    c = (p + z * z / (2 * n)) / d
    h = z * np.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / d
    return c - h, c + h


def boot_ci(x, n=2000):
    x = np.asarray(x)
    means = rng.choice(x, size=(n, len(x)), replace=True).mean(axis=1)
    return np.percentile(means, [2.5, 97.5])


def save(fig, name):
    fig.savefig(os.path.join(VIS, name))
    plt.close(fig)


def main():
    meta = pd.read_csv(os.path.join(DATA, "metadata.csv"))
    img = pd.read_csv(os.path.join(RES, "results_images.csv"))
    att = pd.read_csv(os.path.join(RES, "results_attacks.csv"))
    sw = pd.read_csv(os.path.join(RES, "results_alpha_sweep.csv"))
    ctl = pd.read_csv(os.path.join(RES, "results_controls.csv"))
    summ = json.load(open(os.path.join(RES, "run_summary.json")))
    prim = summ["config"]["primary_protocol"]
    cat = meta.set_index("image_id")[["category", "is_synthetic", "mean_intensity", "contrast_std", "key"]]
    out = {}

    im = img.merge(meta[["image_id", "is_synthetic", "category"]], on="image_id")
    def desc(s):
        return dict(mean=s.mean(), std=s.std(), median=s.median(), min=s.min(), max=s.max())
    out["E1"] = {
        "n_images": int(len(im)),
        "psnr_all": desc(im.psnr_db), "ssim_all": desc(im.ssim),
        "psnr_real": desc(im[im.is_synthetic == 0].psnr_db), "ssim_real": desc(im[im.is_synthetic == 0].ssim),
        "psnr_synthetic": desc(im[im.is_synthetic == 1].psnr_db), "ssim_synthetic": desc(im[im.is_synthetic ==
1].ssim),
        "n_psnr_below_30": int((im.psnr_db < 30).sum()),
        "n_ssim_below_0.98": int((im.ssim < 0.98).sum()),
        "ssim_below_0.98_ids": list(im[im.ssim < 0.98].image_id),
        "psnr_float_mean": float(im.psnr_float_db.mean()),
        "max_abs_diff_max": float(im.max_abs_diff.max()), "mean_abs_diff_mean": float(im.mean_abs_diff.mean()),
        "t_embed_ms_median": float(im.t_embed_ms.median()), "key_mb": float(im.key_bytes.median() / 1e6),
    }
    t1 = pd.DataFrame({"PSNR (dB)": im.psnr_db.describe(), "SSIM": im.ssim.describe()}).round(4)
    t1.to_csv(os.path.join(TAB, "imperceptibility_summary.csv"))

    a = att[att.protocol == prim].merge(cat, left_on="image_id", right_index=True)
    rows = []
    for name in ATTACK_NAMES:
        x = a[a.attack == name]
        lo, hi = boot_ci(x.nc)
        k = int((x.nc >= THR).sum())
        wl, wh = wilson(k, len(x))
        real = x[x.is_synthetic == 0]
        rows.append(dict(attack=name, family=ATTACK_FAMILY[name], n=len(x), mean_nc=x.nc.mean(), ci_lo=lo,
ci_hi=hi,
                         median_nc=x.nc.median(), std_nc=x.nc.std(), min_nc=x.nc.min(), pass_rate=k / len(x),
                         mean_zncc=x.zncc.mean(), median_zncc=x.zncc.median(), zncc_pass_rate=(x.zncc >=
THR).mean(),
                         pass_lo=wl, pass_hi=wh, mean_ber=x.ber.mean(), mean_nc_real_only=real.nc.mean(),
                         pass_rate_real_only=(real.nc >= THR).mean(), t_attack_ms=x.t_attack_ms.median(),
                         t_extract_ms=x.t_extract_ms.median()))
    rob = pd.DataFrame(rows)
    rob.to_csv(os.path.join(TAB, "robustness_by_attack.csv"), index=False)
    fam = a.groupby("family").agg(mean_nc=("nc", "mean"), pass_rate=("nc", lambda s: (s >= THR).mean()),
n=("nc", "size")).reset_index()
    fam.to_csv(os.path.join(TAB, "robustness_by_family.csv"), index=False)

    other = att[att.protocol != prim].merge(cat, left_on="image_id", right_index=True)
    cmp_rows = []
    for name in ATTACK_NAMES:
        u = a[a.attack == name].set_index("image_id").nc
        f = other[other.attack == name].set_index("image_id").nc
        cmp_rows.append(dict(attack=name, mean_nc_uint8=u.mean(), mean_nc_float=f.mean(),
                             max_abs_diff_per_image=(u - f.reindex(u.index)).abs().max()))
```

```
    pd.DataFrame(cmp_rows).to_csv(os.path.join(TAB, "protocol_comparison.csv"), index=False)

    out["E2"] = {
        "protocol": prim, "n_trials": int(len(a)), "mean_nc": float(a.nc.mean()),
        "n_trials_attacks_only": int((a.attack != "No Attack").sum()),
        "n_pass_attacks_only": int(((a.attack != "No Attack") & (a.nc >= THR)).sum()),
        "pass_rate_attacks_only": float((a[a.attack != "No Attack"].nc >= THR).mean()),
        "mean_nc_attacks_only": float(a[a.attack != "No Attack"].nc.mean()),
        "pass_rate": float((a.nc >= THR).mean()), "n_pass": int((a.nc >= THR).sum()),
        "pass_rate_real_only": float((a[a.is_synthetic == 0].nc >= THR).mean()),
        "mean_nc_real_only": float(a[a.is_synthetic == 0].nc.mean()),
        "mean_nc_float_protocol": float(other.nc.mean()), "pass_rate_float_protocol": float((other.nc >=
THR).mean()),
        "mean_ber": float(a.ber.mean()), "mean_zncc": float(a.zncc.mean()), "zncc_pass_rate": float((a.zncc >=
THR).mean()),
        "zncc_clean_mean": float(a[a.attack == "No Attack"].zncc.mean()),
        "clean_nc_mean": float(a[a.attack == "No Attack"].nc.mean()), "clean_nc_min": float(a[a.attack == "No
Attack"].nc.min()),
        "attacks_with_pass_rate_ge_0.95": list(rob[rob.pass_rate >= 0.95].attack),
        "weakest": rob.sort_values("pass_rate").head(3)[["attack", "pass_rate", "mean_nc"]].to_dict("records"),
    }

    piv = a.pivot(index="image_id", columns="attack", values="nc").join(cat)
    cor = []
    for name in ATTACK_NAMES:
        if name == "No Attack":
            continue
        r1 = stats.spearmanr(piv["contrast_std"], piv[name])
        r2 = stats.spearmanr(piv["mean_intensity"], piv[name])
        cor.append(dict(attack=name, rho_contrast=r1[0], p_contrast=r1[1], rho_mean_intensity=r2[0],
p_mean_intensity=r2[1]))
    pd.DataFrame(cor).round(4).to_csv(os.path.join(TAB, "nc_vs_image_statistics.csv"), index=False)

    ct = a.groupby(["category", "family"]).nc.mean().unstack().round(3)
    ct["n_images"] = meta.groupby("category").size()
    ct.to_csv(os.path.join(TAB, "nc_by_category_family.csv"))
    ca = a.groupby(["category", "attack"]).nc.mean().unstack().round(3)
    ca.to_csv(os.path.join(TAB, "nc_by_category_attack.csv"))

    swo = sw[sw.status == "ok"]
    s = swo.groupby(["attack", "alpha"]).agg(psnr=("psnr_db", "mean"), psnr_sd=("psnr_db", "std"),
ssim=("ssim", "mean"),
                                             nc_mean=("nc", "mean"), nc_median=("nc", "median"),
                                             nc_q25=("nc", lambda x: x.quantile(.25)), nc_q75=("nc", lambda x:
x.quantile(.75)),
                                             zncc_mean=("zncc", "mean"), ber_mean=("ber", "mean"),
                                             pass_rate=("nc", lambda x: (x >= THR).mean())).reset_index()
    s.round(4).to_csv(os.path.join(TAB, "alpha_sweep_summary.csv"), index=False)
    out["E3"] = {"table": s.round(4).to_dict("records")}
    logo0 = make_watermark_logo(summ["config"]["logo_size"], summ["config"]["logo_text"])
    U_, S_, Vt_ = np.linalg.svd(logo0)
    rank1 = np.clip((U_[:, :1] * S_[:1]) @ Vt_[:1], 0, 255)
    out["metric_baselines"] = {
        "logo_fill_ratio": float((logo0 > 0).mean()),
        "nc_constant_white_image": normalized_correlation(logo0, np.full_like(logo0, 255.0)),
        "nc_rank1_approximation": normalized_correlation(logo0, rank1),
        "zncc_rank1_approximation": zero_normalized_correlation(logo0, rank1),
        "sigma1_energy_fraction": float(S_[0] ** 2 / (S_ ** 2).sum()),
    }

    def sel(prefix):
        return ctl[ctl.control.str.startswith(prefix)].nc
    g = sel("genuine")
    crow = []
    for lbl, pre in [("genuine: own key, clean watermarked image", "genuine"),
                     ("a: other host, UNwatermarked, key of image i", "a:"),
                     ("b: other host, watermarked, key of image i", "b:"),
                     ("c: same host, different logo embedded, decoded with key A (NC vs logo A)", "c: "),
                     ("c-ref: same host, logo B decoded with its own key (NC vs logo B)", "c-ref"),
                     ("d: own host unwatermarked, own key (degenerate)", "d:")]:
        x = sel(pre)
        z = ctl[ctl.control.str.startswith(pre)].zncc
        crow.append(dict(control=lbl, n=len(x), mean_nc=x.mean(), median=x.median(), p95=x.quantile(.95),
max=x.max(),
                         frac_ge_0_75=(x >= THR).mean(), frac_ge_0_90=(x >= .9).mean(),
                         mean_zncc=z.mean(), p95_zncc=z.quantile(.95), max_zncc=z.max(), frac_zncc_ge_0_75=(z
>= THR).mean()))
    pd.DataFrame(crow).round(4).to_csv(os.path.join(TAB, "controls_summary.csv"), index=False)
    auc = {}
    for k, pre in [("a", "a:"), ("b", "b:"), ("c", "c: ")]:
        x = sel(pre)
        u = stats.mannwhitneyu(g, x, alternative="greater")
        auc[k] = float(u.statistic / (len(g) * len(x)))
```

```
    out["E4"] = {"summary": pd.DataFrame(crow).round(4).to_dict("records"), "auc_genuine_vs_control": auc}

    wrong_p95 = float(pd.concat([sel("a:"), sel("b:")]).quantile(.95))
    out["E4"]["wrong_image_p95"] = wrong_p95
    out["E4"]["attacked_genuine_below_wrong_p95"] = float((a.nc < wrong_p95).mean())

    tim = a.groupby("attack").agg(t_attack_ms=("t_attack_ms", "median"), t_extract_ms=("t_extract_ms",
"median")).reset_index()
    tim.round(3).to_csv(os.path.join(TAB, "timing_by_attack.csv"), index=False)
    out["timing"] = {"embed_ms_median": float(img.t_embed_ms.median()),
                     "extract_ms_median": float(a.t_extract_ms.median()),
                     "throughput_embed_per_s_single_core": float(1000 / img.t_embed_ms.median()),
                     "peak_rss_mb": float(img.peak_rss_mb.max())}

    json.dump(out, open(os.path.join(RES, "analysis_summary.json"), "w"), indent=2, default=float)

    fig, ax = plt.subplots(1, 2, figsize=(9, 3.3))
    real = im[im.is_synthetic == 0]; syn = im[im.is_synthetic == 1]
    ax[0].hist(real.psnr_db, bins=np.arange(35.4, 44.2, 0.2), color=BLUE, label=f"real (n={len(real)})")
    ax[0].hist(syn.psnr_db, bins=np.arange(35.4, 44.2, 0.2), color=ORANGE, label=f"synthetic (n={len(syn)})")
    ax[0].axvline(30, color=INK, lw=1, ls="--"); ax[0].text(30.1, ax[0].get_ylim()[1] * .9, "30 dB",
fontsize=8)
    ax[0].set_xlabel("PSNR of watermarked image (dB)"); ax[0].set_ylabel("images"); ax[0].legend()
    ax[1].hist(real.ssim, bins=np.linspace(0.55, 1.0, 46), color=BLUE)
    ax[1].hist(syn.ssim, bins=np.linspace(0.55, 1.0, 46), color=ORANGE)
    ax[1].axvline(0.98, color=INK, lw=1, ls="--"); ax[1].text(0.9805, ax[1].get_ylim()[1] * .9, "0.98",
fontsize=8)
    ax[1].set_xlabel("SSIM of watermarked image"); ax[1].set_ylabel("images")
    fig.suptitle(f"Imperceptibility at alpha = {summ['config']['alpha']} across {len(im)} images", y=1.02,
fontsize=11)
    fig.tight_layout(); save(fig, "fig_imperceptibility_hist.png")

    order = rob.sort_values("median_nc")["attack"].tolist()
    fig, ax = plt.subplots(figsize=(8.6, 6.2))
    data = [a[a.attack == n].nc.values for n in order]
    bp = ax.boxplot(data, orientation="horizontal", patch_artist=True, widths=0.6, showfliers=True,
                    flierprops=dict(marker="o", markersize=2.5, markerfacecolor=GREY, markeredgecolor="none",
alpha=.6),
                    medianprops=dict(color=INK, lw=1.4), whiskerprops=dict(color=GREY),
capprops=dict(color=GREY),
                    boxprops=dict(edgecolor=GREY))
    for patch, n in zip(bp["boxes"], order):
        patch.set_facecolor(BLUE if rob.set_index("attack").loc[n, "median_nc"] >= THR else ORANGE)
        patch.set_alpha(.85)
    ax.set_yticklabels(order)
    ax.axvline(THR, color=INK, ls="--", lw=1)
    ax.text(THR - .005, len(order) + 0.55, "threshold 0.75", fontsize=8, ha="right", va="center")
    for i, n in enumerate(order, 1):
        pr = rob.set_index("attack").loc[n, "pass_rate"]
        ax.text(1.03, i, f"{pr*100:.0f}%", va="center", fontsize=8.5, color=INK)
    ax.text(1.03, len(order) + 0.55, "NC >= 0.75", fontsize=8, color=INK, va="center")
    ax.set_xlim(-0.02, 1.12); ax.set_xlabel("Normalized correlation (NC) of extracted watermark, 100 images")
    ax.set_title("Robustness per attack: NC over 100 images (box = IQR, line = median)", loc="left")
    ax.grid(axis="y", visible=False)
    fig.tight_layout(); save(fig, "fig_nc_boxplot.png")

    r = rob.sort_values("pass_rate")
    fig, ax = plt.subplots(figsize=(8, 5.4))
    ax.barh(r.attack, r.pass_rate * 100, color=[BLUE if v >= .9 else ORANGE for v in r.pass_rate], height=.62)
    ax.errorbar(r.pass_rate * 100, np.arange(len(r)), xerr=[np.clip((r.pass_rate - r.pass_lo) * 100, 0, None),
np.clip((r.pass_hi - r.pass_rate) * 100, 0, None)],
                fmt="none", ecolor=INK, elinewidth=1, capsize=2)
    for i, v in enumerate(r.pass_rate):
        ax.text(min(v * 100 + 3.5, 104), i, f"{v*100:.0f}%", va="center", fontsize=8.5)
    ax.set_xlim(0, 112); ax.set_xlabel("Images with NC >= 0.75 (%)  - error bars: 95% Wilson interval")
    ax.set_title("Detection pass rate per attack (n = 100 images each)")
    ax.grid(axis="y", visible=False)
    fig.tight_layout(); save(fig, "fig_pass_rate.png")

    fig, ax = plt.subplots(1, 3, figsize=(11.5, 3.5))
    p0 = s[s.attack == "No Attack"]
    ax[0].plot(p0.alpha, p0.psnr, color=BLUE, marker="o", ms=4, lw=1.6)
    ax[0].fill_between(p0.alpha, p0.psnr - p0.psnr_sd, p0.psnr + p0.psnr_sd, color=BLUE, alpha=.15, lw=0)
    ax[0].axhline(30, color=INK, lw=1, ls="--"); ax[0].set_ylabel("mean PSNR (dB)")
    ax[0].set_title("Imperceptibility (band = +-1 sd)")
    cols = {"JPEG Q=50": BLUE, "Gaussian Noise (var=0.02)": ORANGE, "Gaussian Blur (sigma=1.5)": AQUA,
"Rotation 5 deg": GREY}
    for an, c in cols.items():
        q = s[s.attack == an]
        ax[1].plot(q.alpha, q.nc_mean, color=c, marker="o", ms=3.5, lw=1.5, label=an)
        ax[2].plot(q.alpha, q.pass_rate * 100, color=c, marker="o", ms=3.5, lw=1.5, label=an)
    ax[1].axhline(THR, color=INK, lw=1, ls="--"); ax[1].set_ylim(0.4, 1.02); ax[1].set_ylabel("mean NC (100
images)")
```

```
    ax[1].set_title("Robustness: mean NC after attack"); ax[1].legend(fontsize=7.5, loc="lower right")
    ax[2].set_ylim(0, 102); ax[2].set_ylabel("images with NC >= 0.75 (%)"); ax[2].set_title("Robustness:
detection pass rate")
    for x_ in ax:
        x_.set_xlabel("embedding strength alpha")
    fig.tight_layout(); save(fig, "fig_alpha_tradeoff.png")

    fig, ax = plt.subplots(figsize=(8, 5.2))
    rr = rob.sort_values("mean_nc")
    yy = np.arange(len(rr))
    ax.barh(yy + .18, rr.mean_nc, height=.34, color=BLUE, label="NC (original metric)")
    ax.barh(yy - .18, rr.mean_zncc, height=.34, color=ORANGE, label="ZNCC (mean-centred)")
    ax.set_yticks(yy); ax.set_yticklabels(rr.attack); ax.axvline(THR, color=INK, ls="--", lw=1)
    ax.set_xlabel("mean over 100 images"); ax.legend(loc="lower right"); ax.grid(axis="y", visible=False)
    ax.set_title("Plain NC is optimistic: ZNCC removes the brightness offset")
    fig.tight_layout(); save(fig, "fig_nc_vs_zncc.png")

    fig, ax = plt.subplots(figsize=(8, 4))
    bins = np.linspace(0, 1, 41)
    ax.hist(sel("a:"), bins=bins, alpha=.85, color=ORANGE, label="a: other host, unwatermarked", density=True)
    ax.hist(sel("b:"), bins=bins, alpha=.55, color=AQUA, label="b: other host, watermarked (wrong key)",
density=True)
    ax.hist(g, bins=bins, alpha=.75, color=BLUE, label="genuine: own key", density=True)
    ax.axvline(THR, color=INK, ls="--", lw=1)
    ax.set_xlabel("NC against the owner's logo"); ax.set_ylabel("density")
    ax.set_title("Control experiments: wrong-image/wrong-key NC overlaps the 0.75 threshold")
    ax.legend(loc="upper center", fontsize=8)
    fig.tight_layout(); save(fig, "fig_controls_hist.png")

    cc = sel("c: "); cr = sel("c-ref")
    fig, ax = plt.subplots(figsize=(6.4, 3.4))
    vals = [g.mean(), cc.mean(), cr.mean(), sel("a:").mean(), sel("b:").mean()]
    labs = ["genuine\n(logo A, key A)", "logo B embedded,\ndecoded with key A", "logo B embedded,\ndecoded with
key B", "other host,\nunwatermarked", "other host,\nwrong key"]
    ax.bar(labs, vals, color=[BLUE, ORANGE, BLUE, GREY, GREY], width=.6)
    for i, v in enumerate(vals):
        ax.text(i, v + .015, f"{v:.3f}", ha="center", fontsize=9)
    ax.set_ylim(0, 1.12); ax.set_ylabel("mean NC vs the logo tested")
    ax.set_title("False-positive problem: the wrong logo is reported as the owner's logo")
    ax.tick_params(axis="x", labelsize=8); ax.grid(axis="x", visible=False)
    fig.tight_layout(); save(fig, "fig_false_positive.png")

    hm = ct.drop(columns="n_images")
    hm = hm.loc[hm.index != "synthetic_control"]
    fig, ax = plt.subplots(figsize=(8.4, 6.2))
    im_ = ax.imshow(hm.values, cmap="Blues", vmin=0, vmax=1, aspect="auto")
    ax.set_xticks(range(hm.shape[1])); ax.set_xticklabels(hm.columns, rotation=30, ha="right")
    ax.set_yticks(range(hm.shape[0])); ax.set_yticklabels([f"{c} ({int(ct.loc[c,'n_images'])})" for c in
hm.index])
    for i in range(hm.shape[0]):
        for j in range(hm.shape[1]):
            v = hm.values[i, j]
            ax.text(j, i, f"{v:.2f}", ha="center", va="center", fontsize=8, color="white" if v > .62 else INK)
    ax.grid(False); fig.colorbar(im_, ax=ax, fraction=.04, label="mean NC")
    ax.set_title("Mean NC by image category and attack family (real images)")
    fig.tight_layout(); save(fig, "fig_category_heatmap.png")

    fig, ax = plt.subplots(1, 3, figsize=(10.5, 3.3), sharey=True)
    for k, name in enumerate(["Gaussian Noise (var=0.02)", "Rotation 15 deg", "Histogram Equalization"]):
        d = piv
        ax[k].scatter(d["contrast_std"], d[name], s=14, c=np.where(d["is_synthetic"] == 1, ORANGE, BLUE),
alpha=.8, lw=0)
        rho = stats.spearmanr(d["contrast_std"], d[name])[0]
        ax[k].set_title(f"{name}\nSpearman rho = {rho:.2f}", fontsize=9.5)
        ax[k].set_xlabel("image contrast (std of grey levels)"); ax[k].axhline(THR, color=INK, ls="--", lw=1)
    ax[0].set_ylabel("NC after attack")
    fig.tight_layout(); save(fig, "fig_nc_vs_contrast.png")

    fig, ax = plt.subplots(figsize=(6.5, 3.2))
    ax.hist(img.t_embed_ms, bins=20, color=BLUE, label=f"embed (median {img.t_embed_ms.median():.0f} ms)")
    ax.hist(a.t_extract_ms, bins=40, color=ORANGE, alpha=.8, label=f"extract (median
{a.t_extract_ms.median():.0f} ms)")
    ax.set_xlabel("time per 512x512 image (ms, single thread)"); ax.set_ylabel("operations"); ax.legend()
    fig.tight_layout(); save(fig, "fig_timing.png")

    cfg = summ["config"]; alpha = cfg["alpha"]
    logo = make_watermark_logo(cfg["logo_size"], cfg["logo_text"])
    Image.fromarray(logo.astype(np.uint8)).save(os.path.join(VIS, "watermark_logo.png"))
    reps = ["image_066", "image_064", "image_001", "image_051", "image_055", "image_100"]
    fig, axes = plt.subplots(len(reps), 3, figsize=(8.4, 2.8 * len(reps)))
    for r_i, iid in enumerate(reps):
        host = np.asarray(Image.open(os.path.join(DATA, "original", iid + ".png")), dtype=np.float64)
```

```
        w = np.asarray(Image.open(os.path.join(ROOT, "outputs", "watermarked", iid + "_wm.png")),
dtype=np.float64)
        d = np.abs(host - w)
        row = img.set_index("image_id").loc[iid]
        key = meta.set_index("image_id").loc[iid, "key"]
        axes[r_i, 0].imshow(host, cmap="gray", vmin=0, vmax=255); axes[r_i, 0].set_title(f"{iid} ({key})
original", fontsize=8.5)
        axes[r_i, 1].imshow(w, cmap="gray", vmin=0, vmax=255); axes[r_i, 1].set_title(f"watermarked  PSNR
{row.psnr_db:.1f} dB, SSIM {row.ssim:.3f}", fontsize=8.5)
        im2 = axes[r_i, 2].imshow(np.clip(d * 10, 0, 255), cmap="magma", vmin=0, vmax=255); axes[r_i,
2].set_title(f"|difference| x10 (max {d.max():.0f} grey levels)", fontsize=8.5)
        for x in axes[r_i]:
            x.axis("off")
    fig.tight_layout(); save(fig, "fig_representative_comparison.png")

    iid = "image_066"
    host = np.asarray(Image.open(os.path.join(DATA, "original", iid + ".png")), dtype=np.float64)
    w, key = embed_watermark(host, logo, alpha=alpha)
    wq = np.clip(np.rint(w), 0, 255)
    suite = make_attack_suite(cfg["base_seed"] + 100 * (int(iid[-3:]) - 1))
    fig, axes = plt.subplots(3, 6, figsize=(12, 6.6))
    axes[0, 0].imshow(logo, cmap="gray"); axes[0, 0].set_title("original logo", fontsize=8.5)
    for ax_, name in zip(axes.flat[1:], ATTACK_NAMES):
        ext = extract_watermark(suite[name](wq), key)
        ax_.imshow(ext, cmap="gray", vmin=0, vmax=255)
        ax_.set_title(f"{name}\nNC={normalized_correlation(logo, ext):.3f}", fontsize=7.8)
    for ax_ in axes.flat[16:]:
        ax_.axis("off")
    for ax_ in axes.flat:
        ax_.set_xticks([]); ax_.set_yticks([]); ax_.grid(False)
    fig.suptitle("Extracted watermark under each attack (image_066, cameraman)", y=1.0, fontsize=11)
    fig.tight_layout(); save(fig, "fig_extraction_gallery.png")

    T = 128
    sheet = Image.new("L", (10 * T, 10 * T), 255)
    for i, iid in enumerate(meta.image_id):
        sheet.paste(Image.open(os.path.join(ROOT, "outputs", "watermarked", iid + "_wm.png")).resize((T, T)),
((i % 10) * T, (i // 10) * T))
    sheet.save(os.path.join(VIS, "contact_sheet_watermarked.png"))
    print(json.dumps({k: out[k] for k in ("E1", "E2")}, indent=1, default=float)[:3000])


if __name__ == "__main__":
    main()
```

A.10 benchmark_tech.py: Library and scheme benchmarks (S1, S2) (433 lines)

```
from __future__ import annotations

import os
for _v in ("OPENBLAS_NUM_THREADS", "OMP_NUM_THREADS", "MKL_NUM_THREADS"):
    os.environ.setdefault(_v, "1")

import io
import json
import shutil
import subprocess
import sys
import tempfile
import time
import warnings

import cv2
import numpy as np
import pandas as pd
import pywt
import scipy
import scipy.fft
import scipy.linalg
import skimage
from PIL import Image, ImageFilter
from scipy import ndimage
from scipy.fftpack import dct as _dct
from skimage import exposure, filters as skfilters, transform as sktransform
from skimage.metrics import peak_signal_noise_ratio as sk_psnr

from attacks import make_attack_suite
from logo import make_watermark_logo
from metrics import psnr
from watermark_core import dct2, embed_watermark, extract_watermark

warnings.filterwarnings("ignore")
cv2.setNumThreads(1)
```

```
HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, "..", ".."))
RES = os.path.join(ROOT, "results")
VIS = os.path.join(ROOT, "visualizations")
DATA = os.path.join(ROOT, "data")


def load(iid):
    return np.asarray(Image.open(os.path.join(DATA, "original", iid + ".png")), dtype=np.float64)


def tmed(fn, repeats=9, warm=2):
    for _ in range(warm):
        fn()
    ts = []
    for _ in range(repeats):
        t = time.perf_counter()
        fn()
        ts.append((time.perf_counter() - t) * 1e3)
    return float(np.median(ts)), float(np.percentile(ts, 25)), float(np.percentile(ts, 75))


def have(cmd):
    return shutil.which(cmd) is not None


def dct_matrix(n):
    k = np.arange(n)[:, None]; i = np.arange(n)[None, :]
    c = np.sqrt(2.0 / n) * np.cos(np.pi * (2 * i + 1) * k / (2 * n))
    c[0] /= np.sqrt(2)
    return c


def haar_numpy(a):
    a00, a01, a10, a11 = a[0::2, 0::2], a[0::2, 1::2], a[1::2, 0::2], a[1::2, 1::2]
    return (a00 + a01 + a10 + a11) / 2.0


def jpeg_pillow(a, q=50):
    b = io.BytesIO(); Image.fromarray(a.astype(np.uint8)).save(b, "JPEG", quality=q); n = b.tell(); b.seek(0)
    return np.array(Image.open(b)).astype(np.float64), n


def jpeg_cv2(a, q=50):
    ok, buf = cv2.imencode(".jpg", a.astype(np.uint8), [cv2.IMWRITE_JPEG_QUALITY, q])
    return cv2.imdecode(buf, cv2.IMREAD_GRAYSCALE).astype(np.float64), len(buf)


def jpeg_magick(a, q=50, tmp=None):
    src, dst = os.path.join(tmp, "in.png"), os.path.join(tmp, "out.jpg")
    Image.fromarray(a.astype(np.uint8)).save(src)
    subprocess.run(["convert", src, "-quality", str(q), dst], check=True)
    return np.array(Image.open(dst).convert("L")).astype(np.float64), os.path.getsize(dst)


def rot_skimage(a):
    return sktransform.rotate(a / 255.0, 15, resize=False, mode="edge") * 255.0


def rot_cv2(a):
    h, w = a.shape
    M = cv2.getRotationMatrix2D(((w - 1) / 2, (h - 1) / 2), 15, 1.0)
    return cv2.warpAffine(a, M, (w, h), flags=cv2.INTER_LINEAR, borderMode=cv2.BORDER_REPLICATE)


def rot_pillow(a):
    return np.array(Image.fromarray(a.astype(np.uint8)).rotate(15, resample=Image.BILINEAR)).astype(np.float64)


def scale_skimage(a):
    s = sktransform.resize(a, (a.shape[0] // 2, a.shape[1] // 2), anti_aliasing=True, preserve_range=True)
    return sktransform.resize(s, a.shape, anti_aliasing=True, preserve_range=True)


def scale_cv2(a):
    s = cv2.resize(a, (a.shape[1] // 2, a.shape[0] // 2), interpolation=cv2.INTER_AREA)
    return cv2.resize(s, (a.shape[1], a.shape[0]), interpolation=cv2.INTER_LINEAR)


def scale_pillow(a):
    im = Image.fromarray(a.astype(np.uint8))
    s = im.resize((a.shape[1] // 2, a.shape[0] // 2), Image.BILINEAR, reducing_gap=None)
    return np.array(s.resize(im.size, Image.BILINEAR)).astype(np.float64)
```

```
def part_a(images):
    tmp = tempfile.mkdtemp()
    rows = []
    h512 = images[0]
    ll = h512[:256, :256].copy()
    u8 = h512.astype(np.uint8)
    C = dct_matrix(256)

    ops = {
        "2-D DCT-II 256x256": [
            ("scipy.fftpack.dct (project)", lambda: dct2(ll), "ref"),
            ("scipy.fft.dctn", lambda: scipy.fft.dctn(ll, norm="ortho"), ""),
            ("cv2.dct", lambda: cv2.dct(ll), ""),
            ("NumPy matrix C.A.C^T", lambda: C @ ll @ C.T, ""),
        ],
        "SVD 256x256": [
            ("numpy.linalg.svd (project)", lambda: np.linalg.svd(ll, full_matrices=False)[1], "ref"),
            ("scipy.linalg.svd (gesdd)", lambda: scipy.linalg.svd(ll, full_matrices=False, check_finite=False)
[1], ""),
            ("scipy.linalg.svd (gesvd)", lambda: scipy.linalg.svd(ll, full_matrices=False,
lapack_driver="gesvd", check_finite=False)[1], ""),
            ("cv2.SVDecomp", lambda: cv2.SVDecomp(ll)[0].ravel(), ""),
        ],
        "Haar DWT level-1 (LL) 512x512": [
            ("PyWavelets dwt2 (project)", lambda: pywt.dwt2(h512, "haar")[0], "ref"),
            ("NumPy slicing", lambda: haar_numpy(h512), ""),
        ],
        "JPEG round-trip Q=50, 512x512": [
            ("Pillow (project)", lambda: jpeg_pillow(h512)[0], "ref"),
            ("OpenCV imencode/imdecode", lambda: jpeg_cv2(h512)[0], ""),
        ] + ([("ImageMagick convert (process + file I/O)", lambda: jpeg_magick(h512, tmp=tmp)[0], "io")] if
have("convert") else []),
        "Median filter 3x3": [
            ("scipy.ndimage (project)", lambda: ndimage.median_filter(h512, size=3), "ref"),
            ("cv2.medianBlur (uint8)", lambda: cv2.medianBlur(u8, 3), ""),
            ("skimage.filters.median (uint8)", lambda: skfilters.median(u8, np.ones((3, 3), bool)), ""),
        ],
        "Gaussian blur sigma=1.5": [
            ("scipy.ndimage (project)", lambda: ndimage.gaussian_filter(h512, 1.5), "ref"),
            ("cv2.GaussianBlur", lambda: cv2.GaussianBlur(h512, (0, 0), 1.5, borderType=cv2.BORDER_REFLECT),
""),
            ("skimage.filters.gaussian", lambda: skfilters.gaussian(h512, sigma=1.5, preserve_range=True), ""),
        ],
        "Rotation 15 deg (bilinear)": [
            ("skimage.transform.rotate (project)", lambda: rot_skimage(h512), "ref"),
            ("cv2.warpAffine", lambda: rot_cv2(h512), ""),
            ("Pillow Image.rotate", lambda: rot_pillow(h512), ""),
        ],
        "Down/up-scale 0.5x": [
            ("skimage.transform.resize (project)", lambda: scale_skimage(h512), "ref"),
            ("cv2.resize (AREA down, LINEAR up)", lambda: scale_cv2(h512), ""),
            ("Pillow resize (BILINEAR)", lambda: scale_pillow(h512), ""),
        ],
        "Histogram equalisation": [
            ("skimage.exposure.equalize_hist (project)", lambda: exposure.equalize_hist(h512 / 255.0) * 255.0,
"ref"),
            ("cv2.equalizeHist (uint8)", lambda: cv2.equalizeHist(u8).astype(np.float64), ""),
        ],
        "PSNR 512x512": [
            ("skimage.metrics (project)", lambda: sk_psnr(h512, h512 + 1.0, data_range=255), "ref"),
            ("cv2.PSNR", lambda: cv2.PSNR(h512, h512 + 1.0), ""),
            ("NumPy formula", lambda: 10 * np.log10(255 ** 2 / np.mean((h512 - (h512 + 1.0)) ** 2)), ""),
        ],
    }
    for op, variants in ops.items():
        ref = np.asarray(variants[0][1]())
        for label, fn, tag in variants:
            out = np.asarray(fn())
            med, q1, q3 = tmed(fn)
            row = dict(operation=op, implementation=label, median_ms=med, q25_ms=q1, q75_ms=q3,
                       includes_process_and_io=(tag == "io"))
            if out.shape == ref.shape and out.size > 1:
                if op.startswith("Rotation"):
                    sl = (slice(128, 384), slice(128, 384))
                    d = out[sl] - ref[sl]
                    row["agreement"] = "PSNR %.1f dB (interior)" % (10 * np.log10(255 ** 2 / max(np.mean(d **
2), 1e-12)))
                elif op.startswith("SVD") or op.startswith("2-D DCT"):
                    scale = max(np.abs(ref).max(), 1e-12)
                    row["agreement"] = "max rel. diff %.1e" % (np.abs(out - ref).max() / scale)
                elif op.startswith("Haar"):
```

```
                    row["agreement"] = "max abs diff %.1e" % np.abs(out - ref).max()
                else:
                    d = out.astype(np.float64) - ref.astype(np.float64)
                    mse = np.mean(d ** 2)
                    row["agreement"] = "identical" if mse == 0 else "PSNR %.1f dB" % (10 * np.log10(255 ** 2 /
mse))
            else:
                row["agreement"] = "%.4f vs %.4f" % (float(np.ravel(out)[0]), float(np.ravel(ref)[0]))
            rows.append(row)
    j = {"Pillow": jpeg_pillow(h512)[1], "OpenCV": jpeg_cv2(h512)[1]}
    if have("convert"):
        j["ImageMagick"] = jpeg_magick(h512, tmp=tmp)[1]
    shutil.rmtree(tmp, ignore_errors=True)
    return pd.DataFrame(rows), j


NB = 4095


def bits_to_logo(bits):
    b = np.zeros(4096); b[:NB] = np.asarray(bits[:NB], dtype=float)
    return b.reshape(64, 64) * 255.0


class Scheme:
    name = ""
    blind = None
    needs = ""


class OursScheme(Scheme):
    name = "Project: global DWT-DCT-SVD"
    blind = False
    needs = "owner's key (S_orig, Uw, Vw), ~2.1 MB per image"

    def __init__(self, alpha=0.10):
        self.alpha = alpha
        self.logo = make_watermark_logo()

    def embed(self, host):
        w, self.key = embed_watermark(host, self.logo, alpha=self.alpha)
        return w

    def extract(self, img):
        return extract_watermark(img, self.key) > 127.5


class LSBScheme(Scheme):
    name = "Baseline: spatial LSB"
    blind = True
    needs = "payload length only"

    def __init__(self, bits):
        self.bits = np.asarray(bits[:NB], dtype=np.uint8)

    def embed(self, host):
        flat = np.rint(host).astype(np.uint8).flatten()
        flat[:NB] = (flat[:NB] & 0xFE) | self.bits
        return flat.reshape(host.shape).astype(np.float64)

    def extract(self, img):
        flat = np.clip(np.rint(img), 0, 255).astype(np.uint8).flatten()
        return bits_to_logo(flat[:NB] & 1) > 127.5


class BlockDCTScheme(Scheme):
    name = "Baseline: 8x8 block DCT"
    blind = True
    needs = "payload length only"

    def __init__(self, bits, delta=24.0):
        self.bits = np.asarray(bits[:NB], dtype=np.uint8)
        self.delta = delta
        n = 8
        k = np.arange(n)[:, None]; i = np.arange(n)[None, :]
        c = np.sqrt(2.0 / n) * np.cos(np.pi * (2 * i + 1) * k / (2 * n)); c[0] /= np.sqrt(2)
        self.C = c

    def embed(self, host):
        out = host.astype(np.float64).copy()
        H, W = host.shape
        idx = 0
        for by in range(0, H, 8):
```

```
            for bx in range(0, W, 8):
                if idx >= NB:
                    break
                blk = out[by:by + 8, bx:bx + 8]
                d = self.C @ blk @ self.C.T
                a, b = d[4, 3], d[3, 4]
                m = (a + b) / 2
                if self.bits[idx]:
                    d[4, 3], d[3, 4] = m + self.delta / 2, m - self.delta / 2
                else:
                    d[4, 3], d[3, 4] = m - self.delta / 2, m + self.delta / 2
                out[by:by + 8, bx:bx + 8] = self.C.T @ d @ self.C
                idx += 1
        return np.clip(out, 0, 255)

    def extract(self, img):
        H, W = img.shape
        bits = []
        for by in range(0, H, 8):
            for bx in range(0, W, 8):
                if len(bits) >= NB:
                    break
                d = self.C @ img[by:by + 8, bx:bx + 8] @ self.C.T
                bits.append(1 if d[4, 3] > d[3, 4] else 0)
        return bits_to_logo(bits) > 127.5


class BlindWMScheme(Scheme):
    name = "blind-watermark 0.4.4 (block DWT-DCT-SVD)"
    blind = True
    needs = "two integer passwords + payload length"

    def __init__(self, bits):
        from blind_watermark import WaterMark
        self.WaterMark = WaterMark
        self.bits = np.asarray(bits[:NB], dtype=bool)

    def embed(self, host):
        bwm = self.WaterMark(password_img=1, password_wm=1, processes=1)
        bwm.read_img(img=np.dstack([host.astype(np.uint8)] * 3))
        bwm.read_wm(self.bits, mode="bit")
        e = bwm.embed()
        return np.clip(np.rint(e), 0, 255)

    def extract(self, img3):
        bwm = self.WaterMark(password_img=1, password_wm=1, processes=1)
        ex = bwm.extract(embed_img=np.clip(np.rint(img3), 0, 255).astype(np.uint8), wm_shape=NB, mode="bit")
        return bits_to_logo(np.asarray(ex) > 0.5) > 127.5


def channelwise(fn, img):
    if img.ndim == 2:
        return fn(img)
    return np.stack([fn(img[..., c]) for c in range(img.shape[2])], axis=-1)


def luma(img):
    return img if img.ndim == 2 else cv2.cvtColor(img.astype(np.float32),
cv2.COLOR_BGR2GRAY).astype(np.float64)


def part_b(ids, attacks):
    logo = make_watermark_logo()
    bits = (logo.flatten() > 127)[:NB]
    ref_logo = bits_to_logo(bits)
    rows = []
    for k, iid in enumerate(ids):
        host = load(iid)
        suite = make_attack_suite(777 + k)
        schemes = [OursScheme(), BlindWMScheme(bits), LSBScheme(bits), BlockDCTScheme(bits)]
        for sch in schemes:
            try:
                t0 = time.perf_counter(); w = sch.embed(host); t_emb = (time.perf_counter() - t0) * 1e3
                wq = np.clip(np.rint(w), 0, 255)
                p = psnr(host, luma(wq))
            except Exception as exc:
                rows.append(dict(image_id=iid, scheme=sch.name, attack="(embed)", status=f"error: {exc}"))
                continue
            for an in attacks:
                row = dict(image_id=iid, scheme=sch.name, attack=an, psnr_db=p, t_embed_ms=t_emb, status="ok")
                try:
                    att = channelwise(suite[an], wq)
```

```
                    t0 = time.perf_counter(); ex = sch.extract(att); row["t_extract_ms"] = (time.perf_counter()
- t0) * 1e3
                    ex_img = ex.astype(np.float64) * 255.0
                    row["ber"] = float(np.mean((ex_img > 127.5) != (ref_logo > 127.5)))
                    a, b = ref_logo.flatten(), ex_img.flatten()
                    den = np.sqrt((a ** 2).sum()) * np.sqrt((b ** 2).sum())
                    row["nc"] = float((a * b).sum() / den) if den else 0.0
                except Exception as exc:
                    row.update(status=f"error: {type(exc).__name__}: {exc}")
                rows.append(row)
        print("scheme comparison: image", k + 1, "/", len(ids), flush=True)
    return pd.DataFrame(rows)


def tune_delta(ids):
    logo = make_watermark_logo(); bits = (logo.flatten() > 127)[:NB]
    target = np.mean([psnr(load(i), OursScheme().embed(load(i))) for i in ids])
    best = None
    for d in (8, 12, 16, 20, 24, 28, 32, 40, 48):
        p = np.mean([psnr(load(i), np.rint(BlockDCTScheme(bits, d).embed(load(i)))) for i in ids])
        if best is None or abs(p - target) < abs(best[1] - target):
            best = (d, p)
    return best, target


def main():
    os.makedirs(RES, exist_ok=True)
    meta = pd.read_csv(os.path.join(DATA, "metadata.csv"))
    real = meta[meta.is_synthetic == 0].image_id.tolist()
    rng = np.random.default_rng(20260713)

    comp_ids = list(rng.choice(real, 3, replace=False))
    parts = []
    for iid in comp_ids:
        df, jsz = part_a([load(iid)])
        df["image_id"] = iid
        parts.append(df)
    A = pd.concat(parts)
    Agg = A.groupby(["operation", "implementation"], sort=False).agg(
        median_ms=("median_ms", "median"), q25_ms=("q25_ms", "median"), q75_ms=("q75_ms", "median"),
        includes_process_and_io=("includes_process_and_io", "first"), agreement=("agreement",
"first")).reset_index()
    ref = Agg.groupby("operation", sort=False).median_ms.transform("first")
    Agg["speed_vs_project"] = ref / Agg.median_ms
    Agg.round(4).to_csv(os.path.join(RES, "tech_benchmark_components.csv"), index=False)
    print(Agg.round(3).to_string())

    sample = list(rng.choice(real, 30, replace=False))
    tune_ids = sample[:5]
    (delta, p_d), p_target = tune_delta(tune_ids)
    print("block-DCT delta tuned to", delta, "PSNR", round(p_d, 2), "target", round(p_target, 2))
    BlockDCTScheme.__init__.__defaults__ = (delta,)
    attacks = ["No Attack", "JPEG Q=50", "JPEG Q=20", "Gaussian Noise (var=0.02)", "Median Filter 3x3",
               "Gaussian Blur (sigma=1.5)", "Rotation 5 deg", "Cropping 25%", "Scaling 0.5x"]
    B = part_b(sample, attacks)
    B.to_csv(os.path.join(RES, "tech_benchmark_schemes_raw.csv"), index=False)
    ok = B[B.status == "ok"]
    S = ok.groupby(["scheme", "attack"], sort=False).agg(mean_ber=("ber", "mean"), median_ber=("ber",
"median"),
                                                         mean_nc=("nc", "mean"), psnr_db=("psnr_db", "mean"),
                                                         t_embed_ms=("t_embed_ms", "median"),
t_extract_ms=("t_extract_ms", "median"),
                                                         ber_le_0_10=("ber", lambda x: (x <= 0.10).mean()),
n=("ber", "size")).reset_index()
    S.round(4).to_csv(os.path.join(RES, "tech_benchmark_schemes.csv"), index=False)
    print(S.round(3).to_string())

    env = {"python": sys.version.split()[0], "numpy": np.__version__, "scipy": scipy.__version__, "scikit-
image": skimage.__version__,
           "opencv": cv2.__version__, "Pillow": __import__("PIL").__version__, "PyWavelets": pywt.__version__,
           "blind-watermark": __import__("importlib.metadata", fromlist=["version"]).version("blind-
watermark"),
           "ImageMagick": (subprocess.run(["convert", "-version"], capture_output=True,
text=True).stdout.splitlines() or [""])[0] if have("convert") else None,
           "ffmpeg": (subprocess.run(["ffmpeg", "-version"], capture_output=True,
text=True).stdout.splitlines() or [""])[0] if have("ffmpeg") else None,
           "cpu_threads_used": 1, "repeats": 9, "warmup": 2, "component_images": comp_ids, "scheme_images":
sample,
           "jpeg_bytes_example": jsz, "blockdct_delta": delta, "blockdct_psnr": p_d, "project_psnr_tune":
p_target}
    json.dump(env, open(os.path.join(RES, "tech_summary.json"), "w"), indent=2, default=str)
```

```
if __name__ == "__main__":
    main()
```

A.11 video_demo.py: Exploratory per-frame video test (V1) (149 lines)

```
from __future__ import annotations

import os
for _v in ("OPENBLAS_NUM_THREADS", "OMP_NUM_THREADS", "MKL_NUM_THREADS"):
    os.environ.setdefault(_v, "1")

import argparse
import json
import subprocess
import tempfile
import time
import urllib.request

import cv2
import numpy as np
import pandas as pd
from PIL import Image

from logo import make_watermark_logo
from metrics import bit_error_rate, normalized_correlation, psnr
from watermark_core import embed_watermark, extract_watermark

HERE = os.path.dirname(os.path.abspath(__file__))
ROOT = os.path.abspath(os.path.join(HERE, "..", ".."))
RES = os.path.join(ROOT, "results")
CACHE = os.path.join(ROOT, "data", "source_videos")
CLIPS = {
    "vtest.avi": "opencv/opencv/4.x/samples/data/vtest.avi",
    "Megamind.avi": "opencv/opencv/4.x/samples/data/Megamind.avi",
}
SIZE = 512
CODECS = {
    "lossless (FFV1)":  ["-c:v", "ffv1"],
    "H.264 CRF 18":     ["-c:v", "libx264", "-crf", "18", "-pix_fmt", "yuv420p"],
    "H.264 CRF 23":     ["-c:v", "libx264", "-crf", "23", "-pix_fmt", "yuv420p"],
    "H.264 CRF 28":     ["-c:v", "libx264", "-crf", "28", "-pix_fmt", "yuv420p"],
    "MJPEG q=5":        ["-c:v", "mjpeg", "-q:v", "5", "-pix_fmt", "yuvj420p"],
    "MJPEG q=15":       ["-c:v", "mjpeg", "-q:v", "15", "-pix_fmt", "yuvj420p"],
}


def fetch(name):
    os.makedirs(CACHE, exist_ok=True)
    p = os.path.join(CACHE, name)
    if not os.path.exists(p):
        urllib.request.urlretrieve("https://raw.githubusercontent.com/" + CLIPS[name], p)
    return p


def gray_square(frame_bgr):
    g = cv2.cvtColor(frame_bgr, cv2.COLOR_BGR2GRAY)
    s = min(g.shape)
    y0, x0 = (g.shape[0] - s) // 2, (g.shape[1] - s) // 2
    g = g[y0:y0 + s, x0:x0 + s]
    return np.asarray(Image.fromarray(g).resize((SIZE, SIZE), Image.LANCZOS), dtype=np.float64)


def read_frames(path, n):
    cap = cv2.VideoCapture(path)
    if not cap.isOpened():
        raise IOError(f"cannot open {path}")
    fps = cap.get(cv2.CAP_PROP_FPS)
    total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    w, h = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH)), int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    frames = []
    try:
        while len(frames) < n:
            ok, f = cap.read()
            if not ok:
                break
            frames.append(gray_square(f))
    finally:
        cap.release()
    return frames, dict(fps=fps, total_frames=total, width=w, height=h)


def encode_decode(frames, codec_args, fps, tmp):
```

```
    out = os.path.join(tmp, "clip." + ("mkv" if "ffv1" in codec_args else ("avi" if "mjpeg" in codec_args else
"mp4")))
    raw = b"".join(np.clip(np.rint(f), 0, 255).astype(np.uint8).tobytes() for f in frames)
    enc = subprocess.run(["ffmpeg", "-v", "error", "-y", "-f", "rawvideo", "-pix_fmt", "gray", "-s",
f"{SIZE}x{SIZE}",
                          "-r", str(fps), "-i", "-", *codec_args, out], input=raw, capture_output=True)
    if enc.returncode != 0:
        raise RuntimeError(enc.stderr.decode()[:300])
    dec = subprocess.run(["ffmpeg", "-v", "error", "-i", out, "-f", "rawvideo", "-pix_fmt", "gray", "-"],
capture_output=True)
    if dec.returncode != 0:
        raise RuntimeError(dec.stderr.decode()[:300])
    buf = np.frombuffer(dec.stdout, dtype=np.uint8)
    n = buf.size // (SIZE * SIZE)
    decoded = [buf[i * SIZE * SIZE:(i + 1) * SIZE * SIZE].reshape(SIZE, SIZE).astype(np.float64) for i in
range(n)]
    return decoded, os.path.getsize(out)


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("--frames", type=int, default=60)
    args = ap.parse_args()
    logo = make_watermark_logo()
    rows, meta = [], {}
    for clip in CLIPS:
        frames, info = read_frames(fetch(clip), args.frames)
        meta[clip] = dict(info, frames_read=len(frames))
        t0 = time.perf_counter()
        marked, keys = [], []
        for f in frames:
            w, k = embed_watermark(f, logo, alpha=0.10)
            k = {kk: vv for kk, vv in k.items() if kk not in ("U", "Vt")}
            marked.append(np.clip(np.rint(w), 0, 255)); keys.append(k)
        t_embed = (time.perf_counter() - t0) / len(frames)
        fps = info["fps"] if info["fps"] and info["fps"] > 0 else 25
        with tempfile.TemporaryDirectory() as tmp:
            for codec, args_ in CODECS.items():
                row = dict(clip=clip, codec=codec, frames_in=len(frames), status="ok")
                try:
                    dec_wm, size_wm = encode_decode(marked, args_, fps, tmp)
                    dec_ref, size_ref = encode_decode(frames, args_, fps, tmp)
                    n = min(len(dec_wm), len(frames))
                    row["frames_decoded"] = len(dec_wm)
                    t0 = time.perf_counter()
                    ncs, bers = [], []
                    for i in range(n):
                        ex = extract_watermark(dec_wm[i], keys[i])
                        ncs.append(normalized_correlation(logo, ex)); bers.append(bit_error_rate(logo, ex))
                    row["t_extract_ms_per_frame"] = (time.perf_counter() - t0) / n * 1e3
                    row["psnr_wm_vs_orig_db"] = float(np.mean([psnr(frames[i], dec_wm[i]) for i in range(n)]))
                    row["psnr_codec_only_db"] = float(np.mean([psnr(frames[i], dec_ref[i]) for i in range(n)]))
                    row["nc_mean"] = float(np.mean(ncs)); row["nc_min"] = float(np.min(ncs))
                    row["frames_nc_ge_0_75"] = int(np.sum(np.array(ncs) >= 0.75)); row["ber_mean"] =
float(np.mean(bers))
                    row["size_wm_kb"] = size_wm / 1024; row["size_ref_kb"] = size_ref / 1024
                    if codec == "lossless (FFV1)":
                        keep = [i for i in range(len(dec_wm)) if i % 10 != 9]
                        naive = [normalized_correlation(logo, extract_watermark(dec_wm[j], keys[i])) for i, j
in enumerate(keep)]
                        aware = [normalized_correlation(logo, extract_watermark(dec_wm[j], keys[j])) for j in
keep]
                        row["framedrop_nc_naive_index"] = float(np.mean(naive))
                        row["framedrop_nc_with_frame_ids"] = float(np.mean(aware))
                except Exception as exc:
                    row.update(status=f"error: {type(exc).__name__}: {exc}")
                rows.append(row)
        meta[clip]["embed_ms_per_frame"] = t_embed * 1e3
        meta[clip]["embed_fps_single_thread"] = 1 / t_embed
        print(clip, meta[clip], flush=True)
    df = pd.DataFrame(rows)
    df.to_csv(os.path.join(RES, "video_demo_results.csv"), index=False)
    json.dump(meta, open(os.path.join(RES, "video_demo_meta.json"), "w"), indent=2)
    pd.set_option("display.width", 250)
    print(df.round(3).to_string())


if __name__ == "__main__":
    main()
```

A.12 make_diagram.py: Architecture diagrams (98 lines)

```
import matplotlib
```

```
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.patches import FancyBboxPatch, FancyArrowPatch
import os

RESULTS_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "..", "..", "visualizations")


def box(ax, xy, w, h, text, fc="#eaf2f8", ec="#2874a6", fontsize=9.5):
    x, y = xy
    p = FancyBboxPatch((x, y), w, h, boxstyle="round,pad=0.02,rounding_size=0.02",
                       fc=fc, ec=ec, linewidth=1.6)
    ax.add_patch(p)
    ax.text(x + w / 2, y + h / 2, text, ha="center", va="center",
            fontsize=fontsize, wrap=True)


def arrow(ax, p1, p2, color="#333333"):
    a = FancyArrowPatch(p1, p2, arrowstyle="-|>", mutation_scale=14,
                        color=color, linewidth=1.5)
    ax.add_patch(a)


def embedding_diagram():
    fig, ax = plt.subplots(figsize=(11, 4.2))
    ax.set_xlim(0, 11); ax.set_ylim(0, 4.2); ax.axis("off")
    ax.set_title("Watermark Embedding Pipeline", fontsize=13, fontweight="bold", pad=16)

    steps = [
        "Host Image\n(512x512, grayscale)",
        "2D DWT\n(Haar, 1-level)\n-> LL sub-band",
        "2-D DCT\nof LL sub-band\n(global)",
        "SVD\nA = U.S.V^T",
        "S' = S + alpha*Sw\n(Sw: singular\nvalues of logo)",
        "Inverse DCT\n+ Inverse DWT",
        "Watermarked\nImage",
    ]
    n = len(steps)
    w, h = 1.35, 1.5
    gap = (11 - n * w) / (n + 1)
    x = gap
    y = 1.1
    centers = []
    for s in steps:
        box(ax, (x, y), w, h, s, fontsize=9.2)
        centers.append((x, x + w))
        x += w + gap
    for i in range(n - 1):
        arrow(ax, (centers[i][1], y + h / 2), (centers[i + 1][0], y + h / 2))

    box(ax, (centers[4][0] - 0.1, y + h + 0.55), 1.5, 0.85, "Watermark\nLogo (64x64)", fc="#fdebd0",
ec="#b9770e", fontsize=9.2)
    arrow(ax, (centers[4][0] + 0.6, y + h + 0.55), (centers[4][0] + 0.6, y + h), color="#b9770e")

    plt.tight_layout()
    plt.savefig(os.path.join(RESULTS_DIR, "diagram_embedding.png"), dpi=170)
    plt.close(fig)


def extraction_diagram():
    fig, ax = plt.subplots(figsize=(11, 4.0))
    ax.set_xlim(0, 11); ax.set_ylim(0, 4.0); ax.axis("off")
    ax.set_title("Watermark Extraction Pipeline (non-blind, key-based)", fontsize=13, fontweight="bold",
pad=16)

    steps = [
        "Received Image\n(possibly attacked:\nJPEG / noise / crop /\nrotate / scale ...)",
        "2D DWT\n-> LL sub-band",
        "2-D DCT\n+ SVD -> S_att",
        "Sw' = (S_att - S_orig)\n/ alpha",
        "Reconstruct:\nUw . Sw' . Vw^T",
        "Extracted\nWatermark",
    ]
    n = len(steps)
    w, h = 1.55, 1.5
    gap = (11 - n * w) / (n + 1)
    x = gap
    y = 0.9
    centers = []
    for s in steps:
        box(ax, (x, y), w, h, s, fontsize=9.2, fc="#eafaf1", ec="#1e8449")
        centers.append((x, x + w))
        x += w + gap
```

```
    for i in range(n - 1):
        arrow(ax, (centers[i][1], y + h / 2), (centers[i + 1][0], y + h / 2))

    box(ax, (centers[3][0] - 0.3, y + h + 0.5), 2.0, 0.75,
        "Embedding Key\n{S_orig, Uw, Vw, alpha, shapes}", fc="#f9ebea", ec="#943126", fontsize=8.6)
    arrow(ax, (centers[3][0] + 0.7, y + h + 0.5), (centers[3][0] + 0.7, y + h), color="#943126")

    plt.tight_layout()
    plt.savefig(os.path.join(RESULTS_DIR, "diagram_extraction.png"), dpi=170)
    plt.close(fig)


if __name__ == "__main__":
    embedding_diagram()
    extraction_diagram()
    print("done")
```